NATIONAL UNIVERSITY OF SINGAPORE (NUS)

Which Asset Wins Next Month?

Momentum-Weighted Risk Parity for Dynamic Cross-Asset Allocation

FE5209 Financial Econometrics
Group Project

Presentation date: 2 November

[Member 1] [Member 2] [Member 3] [Member 4] [Member 5]

01

The Investment Problem

Capture trends while controlling portfolio concentration

Speaker 1

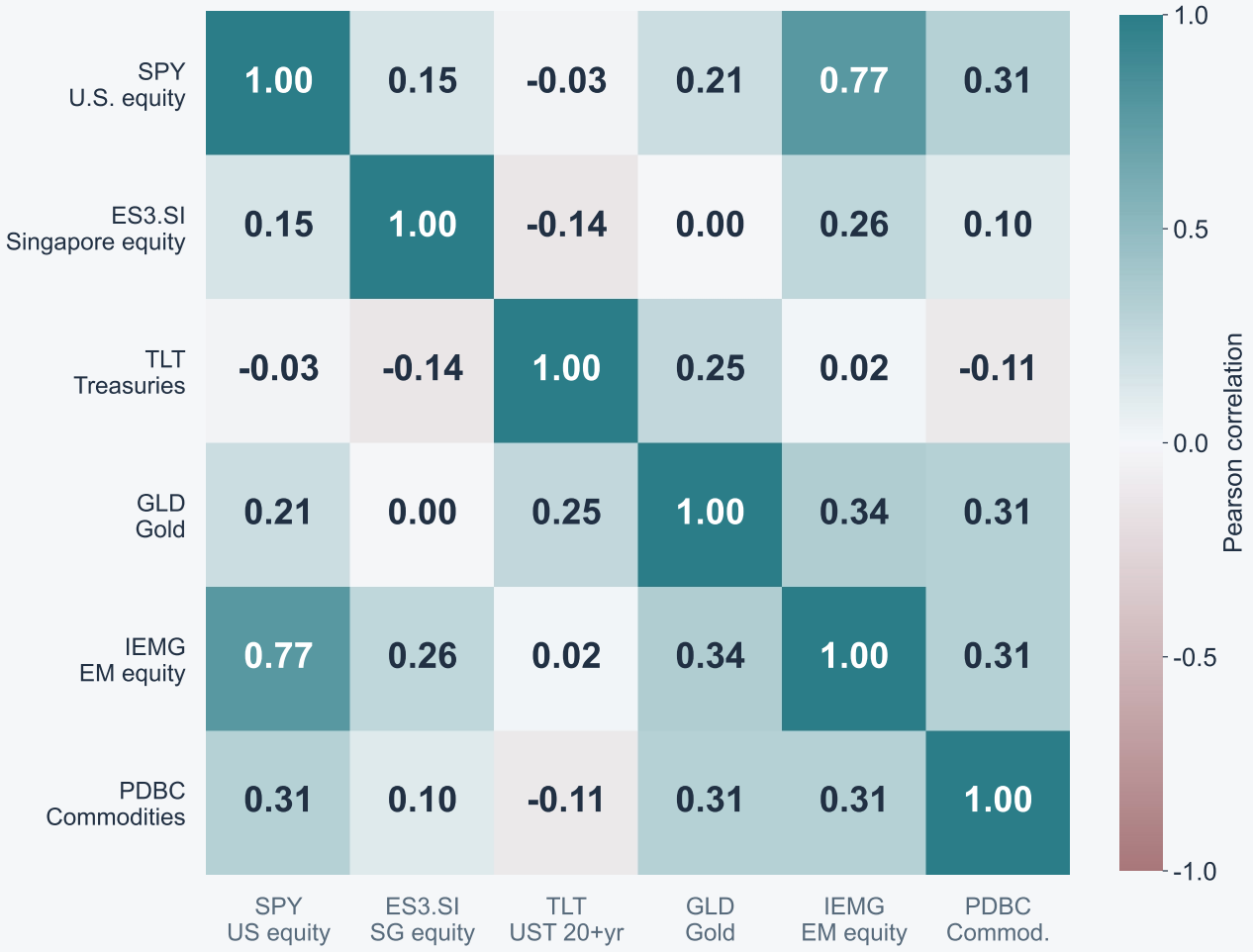
Six risk exposures and one defensive asset

Risk exposure	ETF ticker	Portfolio role
US equity	SPY	Risk asset
Singapore equity	ES3.SI	Risk asset
Long-term US Treasuries	TLT	Risk asset
Gold	GLD	Risk asset
Emerging markets equity	IEMG	Risk asset
Broad commodities	PDBC	Risk asset
Treasury bills	**BIL**	**Defensive asset**

ETF = exchange-traded fund. SGD = Singapore dollar. USD = United States dollar.

BIL retains USD/SGD currency risk. It is not risk-free SGD cash.

Different return relationships support diversification



2020 Jan-2026 Sep / SGD daily total returns / N = 1,742 valuation days

Diversification opportunities

Equity, rate and currency risks can overlap.

Pearson correlation
2020 Jan–2026 Sep

Momentum selection and equal-risk allocation

Dual Momentum

Ranks six-month asset returns

Allocates two 50% capital slots

Emphasizes strong winners

Equal Risk Contribution (ERC)

Balances forecast risk shares

Uses volatility and co-movement

Does not rank momentum strength

Momentum-Gated Risk Parity (MGRP) adds eligibility to equal-risk allocation.

The portfolio-construction question

Can momentum strength improve risk budgeting without sacrificing too much return?

01 **Eligibility**

Which assets qualify?

02 **Risk preferences**

How much forecast risk should each receive?

03 **Capital allocation**

How much money implements those preferences?

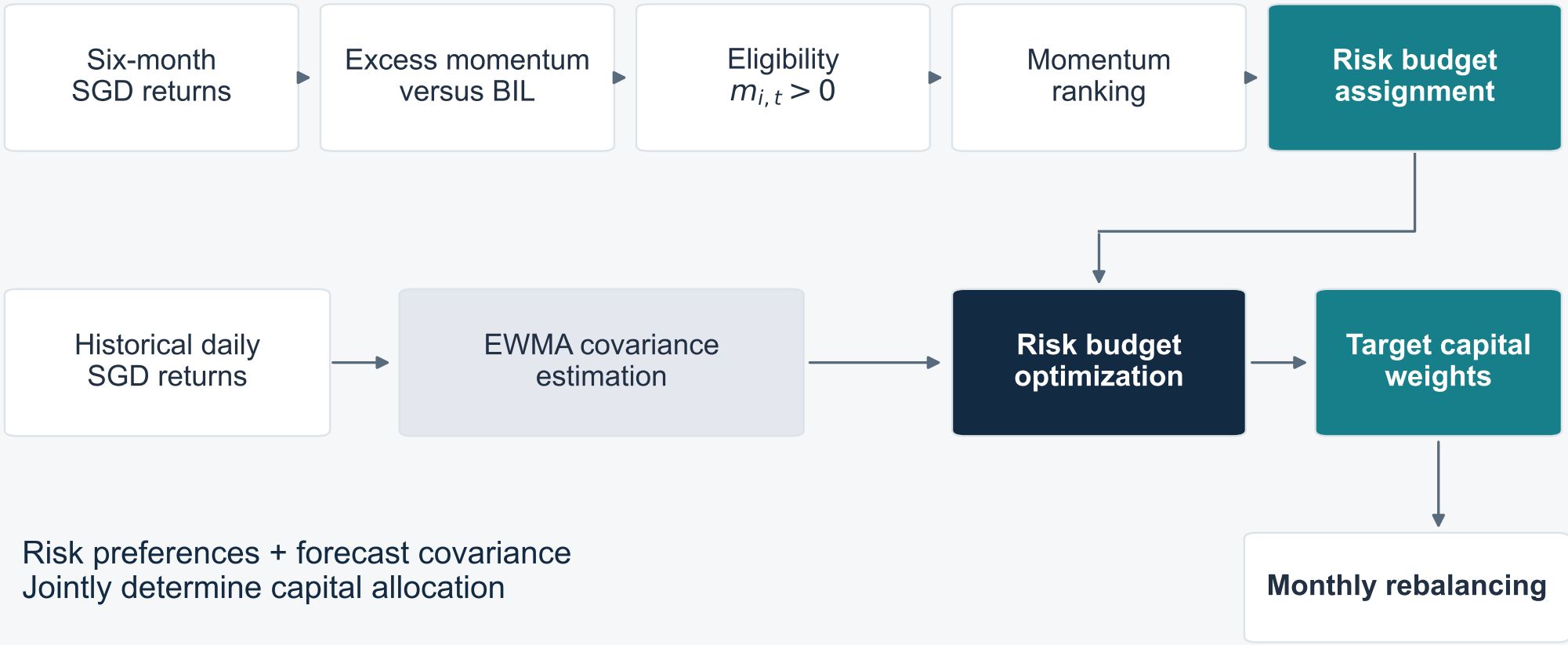
02

From Momentum to Portfolio Weights

Eligibility, risk preferences and covariance-based capital allocation

Speaker 2

Momentum sets risk preferences, covariance sets capital



Risk preferences + forecast covariance
Jointly determine capital allocation

EWMA: Exponentially Weighted Moving Average / BIL: defensive Treasury-bill fund / Schematic, not a statistical result

An asset must beat BIL to qualify

Six-month excess momentum

$$m_{i,t} = \left(\frac{P^{\mathrm{SGD,TR}}_{i,t}}{P^{\mathrm{SGD,TR}}_{i,t-6}} - 1 \right) - \left(\frac{P^{\mathrm{SGD,TR}}_{\mathrm{BIL},t}}{P^{\mathrm{SGD,TR}}_{\mathrm{BIL},t-6}} - 1 \right)$$

Eligibility and the number of qualifying risk assets

$$E_{i,t} = \mathbf{1}\{m_{i,t} > 0\}\,, \qquad k_t = \sum_{i=1}^{6} E_{i,t}$$

i: risk asset. t: month-end. P: SGD total-return index. k: eligible count.

Strictly positive spread. No eligible assets: 100% BIL. One eligible asset: 100% capital.

Stronger momentum receives a larger risk budget

$$b_{i,t} = \frac{k_t - r_{i,t} + 1}{k_t(k_t+1)/2} \quad (i \in \mathcal{E}_t), \qquad b_{i,t} = 0 \quad (i \notin \mathcal{E}_t)$$

k = eligible count. r = rank among eligible assets. Rank 1 is strongest.

Eligible rank	1	2	3	4
Target risk share	40%	30%	20%	10%

Risk budget is not capital weight

Fixed linear rank rule. Budgets sum to 100%. The four-asset row illustrates the rule.

Recent observations receive more weight in covariance

Exponentially Weighted Moving Average (EWMA)

$$Q_d = \lambda Q_{d-1} + (1-\lambda)\varepsilon_d \varepsilon_d^\top, \quad \varepsilon_d = R_d - \bar{R}_{d-1}, \quad \lambda = 0.94,$$

$$\widehat{\Sigma}_d = 0.99 Q_d + 0.01\,\mathrm{diag}(Q_d) + D_d$$

01 **Recent information**

lambda = 0.94. Use the prior expanding mean for innovations.

02 **Fixed stabilization**

Initialize with 252 returns. Apply 1% diagonal shrinkage and a 1e-12 floor.

Q: raw covariance. R: daily asset returns. D: diagonal floor correction. No parameter search.

The optimizer converts risk budgets into capital weights

$$x^* = \underset{x_i>0,\ i\in\mathcal{E}_t}{\arg\min}\left\{\tfrac{1}{2}x^\top\widehat{\Sigma}_t x - \sum\nolimits_{i\in\mathcal{E}_t} b_{i,t}\log x_i\right\},$$

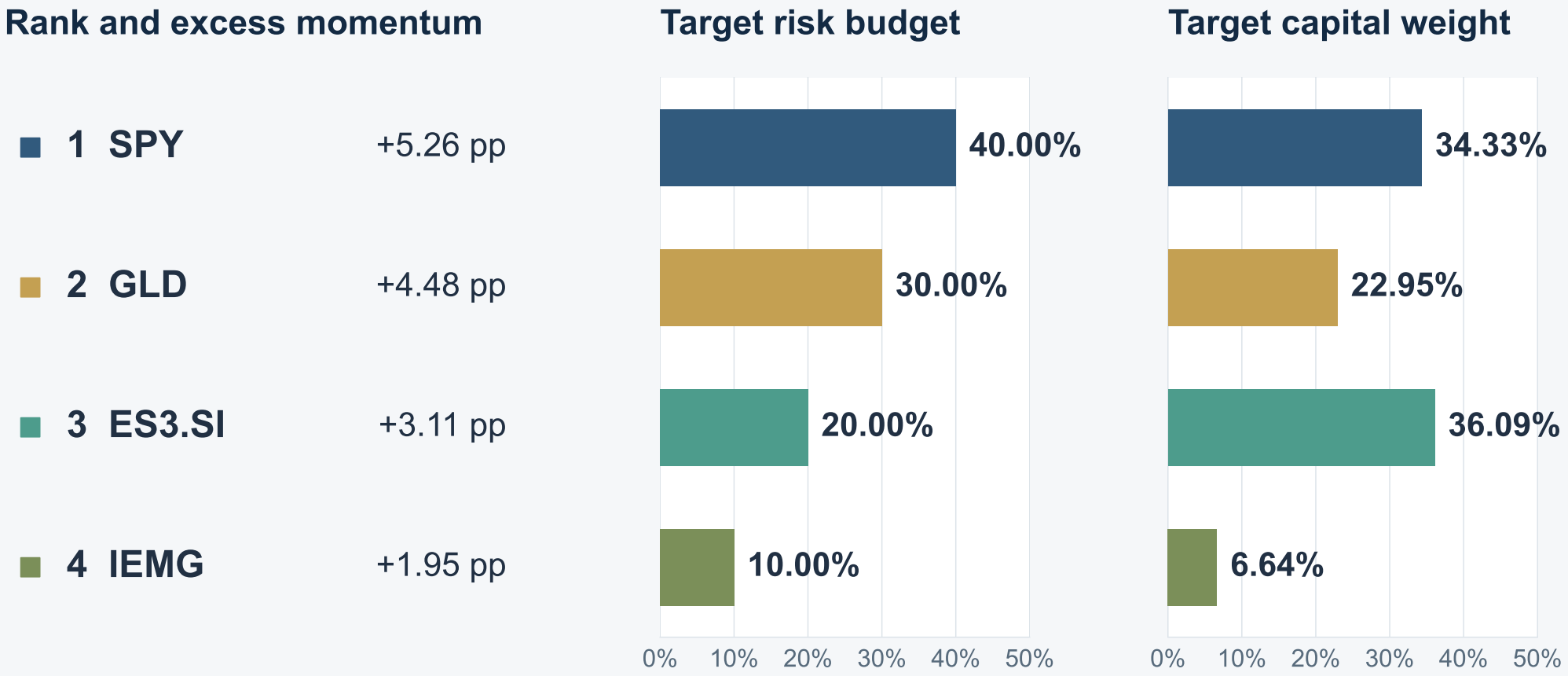
$$w_i = x_i^* \Big/ \sum\nolimits_{j\in\mathcal{E}_t} x_j^*$$

x: positive optimization variable. w: normalized capital weights. b: assigned risk shares.

$$\mathrm{RC}_{i,t}^{\sigma} = \frac{w_i(\widehat{\Sigma}_t w)_i}{\sqrt{w^\top\widehat{\Sigma}_t w}}, \qquad \frac{\mathrm{RC}_{i,t}^{\sigma}}{\sigma_p} = \frac{w_i(\widehat{\Sigma}_t w)_i}{w^\top\widehat{\Sigma}_t w} = b_{i,t}$$

Estimated risk shares equal assigned budgets. This equality is ex ante, not a future guarantee.

Risk budgets and capital differ in a real decision



Risk budget ≠ capital weight

Signal: Dec 2023 / Information through 29 Dec / Execution: 3 Jan 2024 / Excluded: TLT and PDBC

03

Econometric Framework & Validation

Test information timing, covariance forecasts and risk contributions

Speaker 3

Benchmarks separate selection from allocation

Strategy	Eligibility / selection	Allocation	Research role
Equal Weight	All six risk assets	Equal capital: 1/6 each	Passive allocation
EWMA Risk Parity (ERC)	All six risk assets	Equal estimated risk contribution	Risk allocation
Dual Momentum	Top 2 six-month returns; each slot must beat BIL	Two 50% capital slots; failed slot enters BIL	Return-oriented momentum
MGRP	All assets beating BIL	Equal risk budgets among eligible assets	Equal-risk ablation
MWRP	Same eligibility as MGRP	Linear momentum-rank risk budgets; same EWMA covariance	Proposed strategy
Gated Equal Weight	All assets beating BIL	Equal capital among eligible assets	Supplementary ablation

For gated portfolios: no eligible risk asset means 100% BIL / Long-only, fully invested, unlevered / Monthly rebalancing

SGD total returns use adjusted prices and exchange rates

01 **Archived adjusted prices**

Distributions and splits are reflected in total-return indices.

02 **One investor currency**

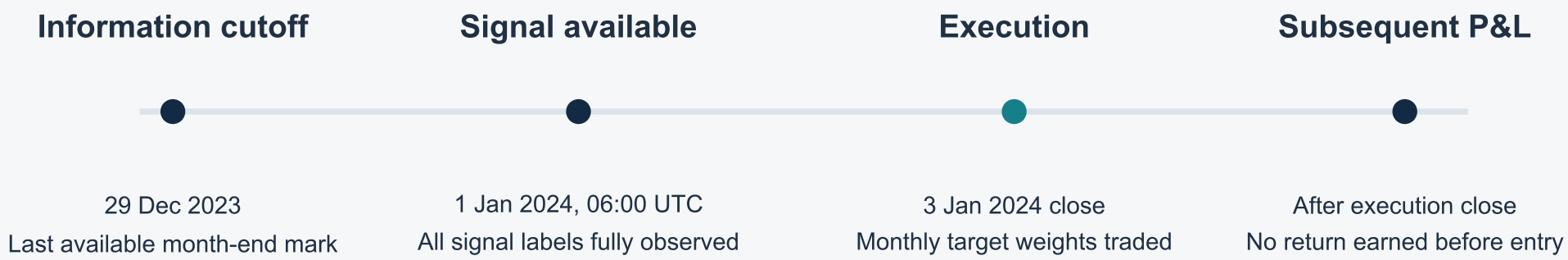
Translate USD funds using SGD per USD. ES3 remains in SGD.

03 **Cross-market valuation**

Use information available at the time. Executions require fresh basket quotes.

FX means foreign exchange. Fund expenses are already reflected in prices, without duplicate deduction.

New positions earn returns only after execution



Identical execution clock and valuation rules across strategies

Concrete frozen example from the first main-evaluation decision / Asynchronous market and FX marks follow the original valuation grid

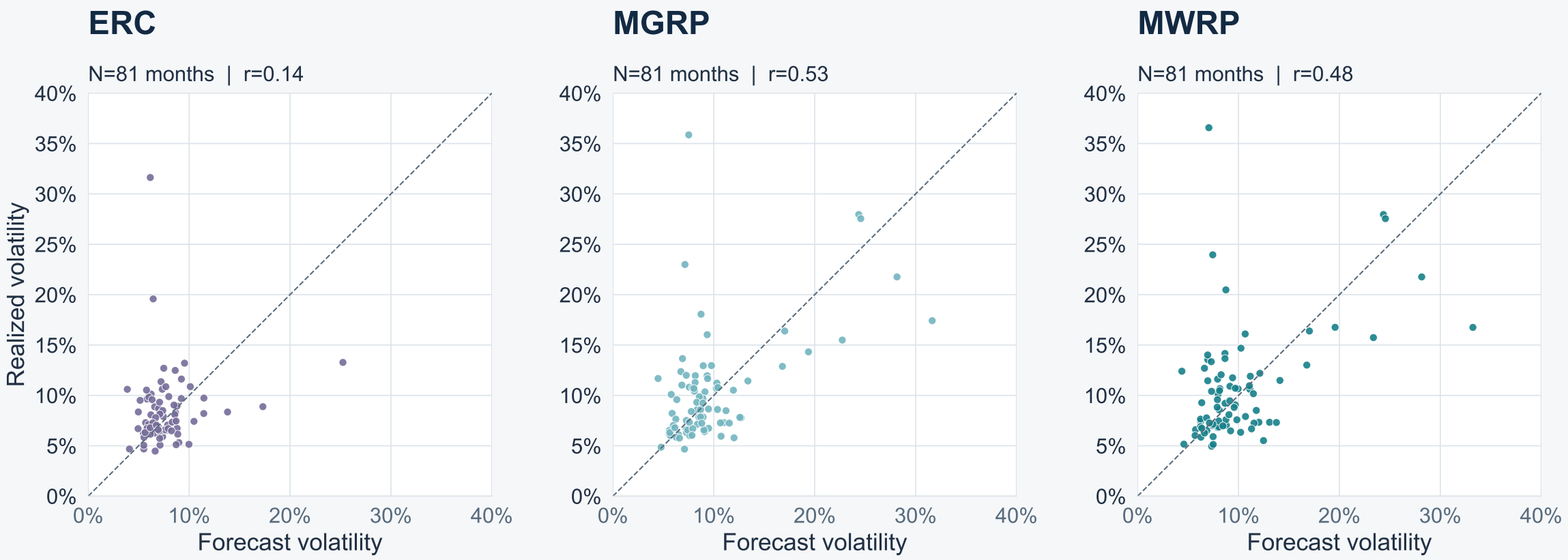
Historical windows test different market environments

Window	Role	Scope
2014–2019	Data preparation and initialization	Common asset history from Nov 2014
2020–2023	Historical validation	48 months, 1,031 portfolio returns
2024 Jan–2026 Sep	Historical evaluation	33 months, 710 portfolio returns
2020 Jan–2026 Sep	Continuous combined history	81 months, 1,741 portfolio returns

Exploratory historical research

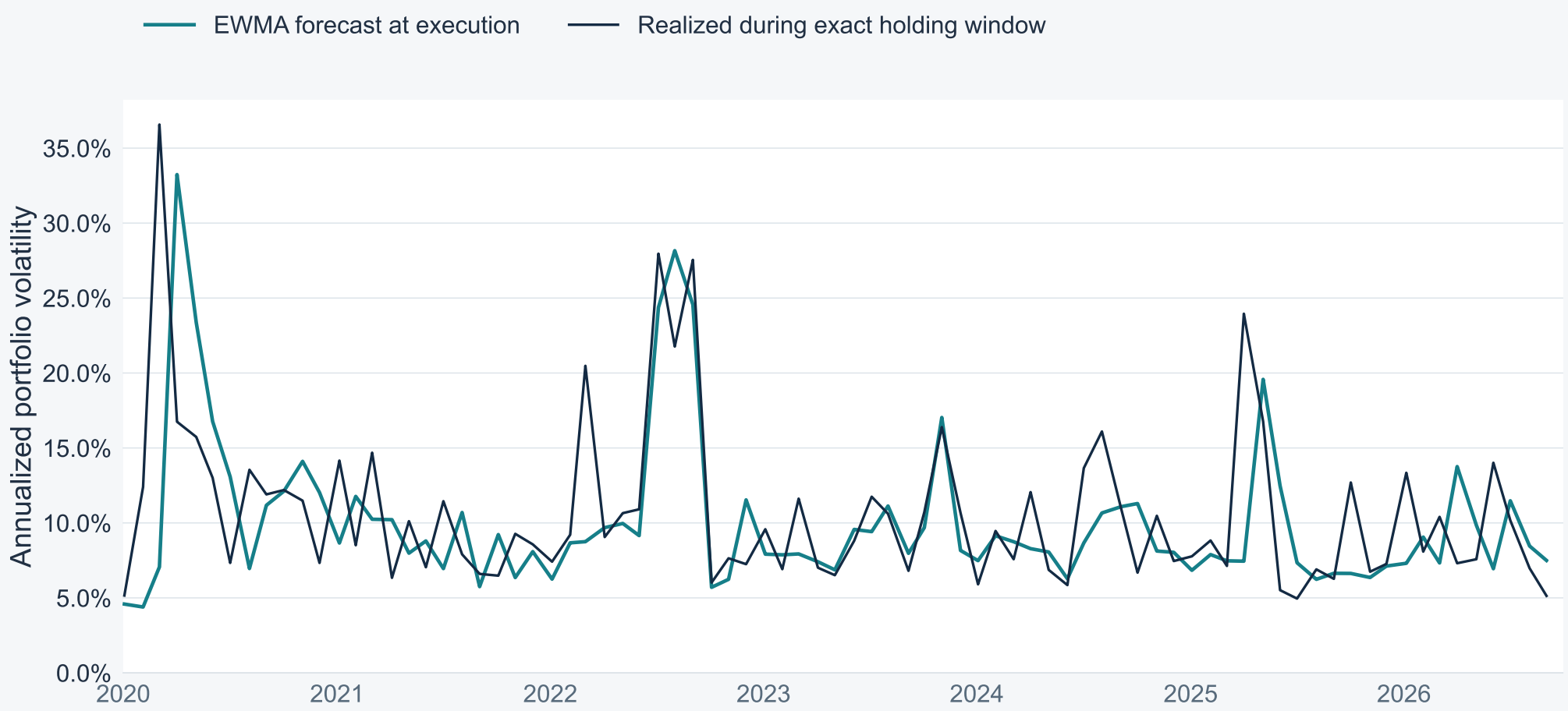
Rules were developed after historical inspection. No untouched prospective claim.

Covariance forecasts provide imperfect risk information



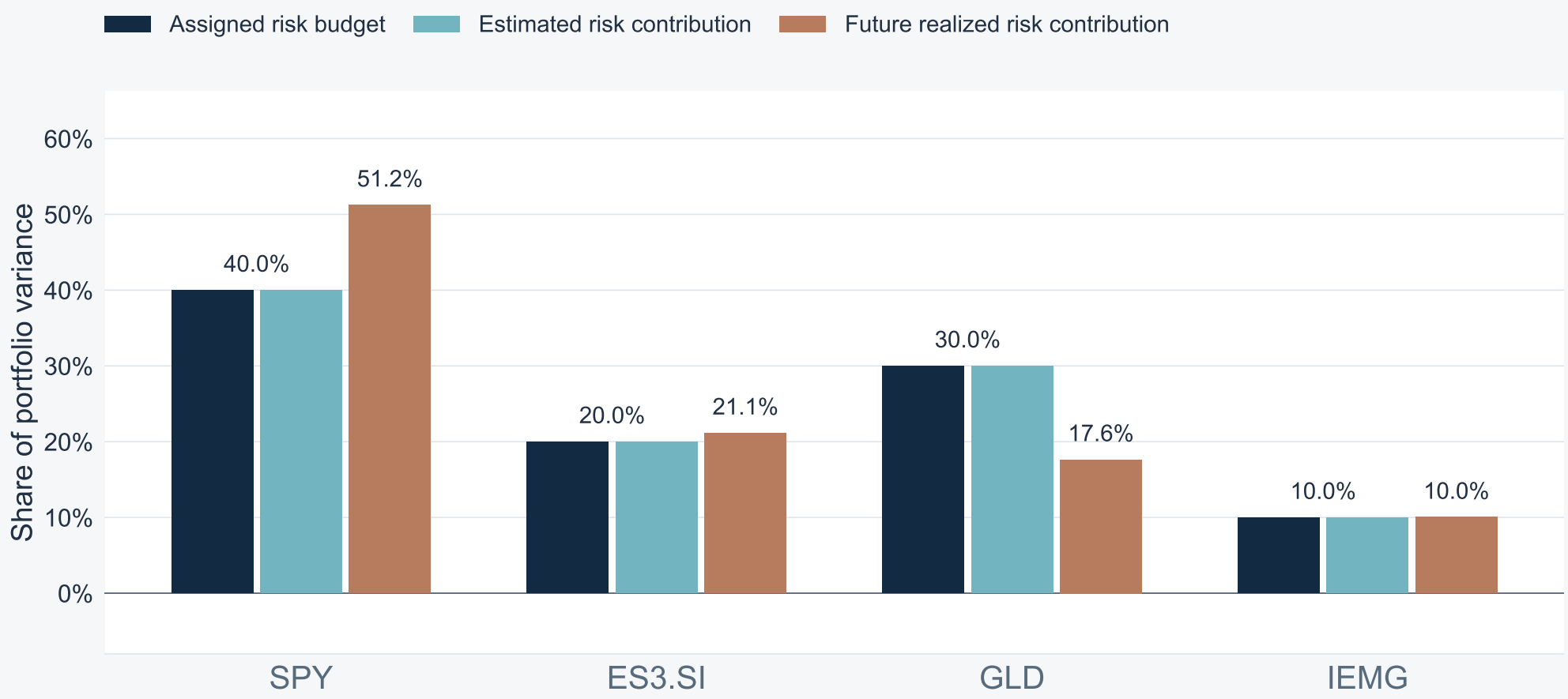
Exact execution-to-next-execution holding windows / One observation per monthly portfolio cohort / Dashed line: forecast = realized

Forecast risk does not track every future risk change



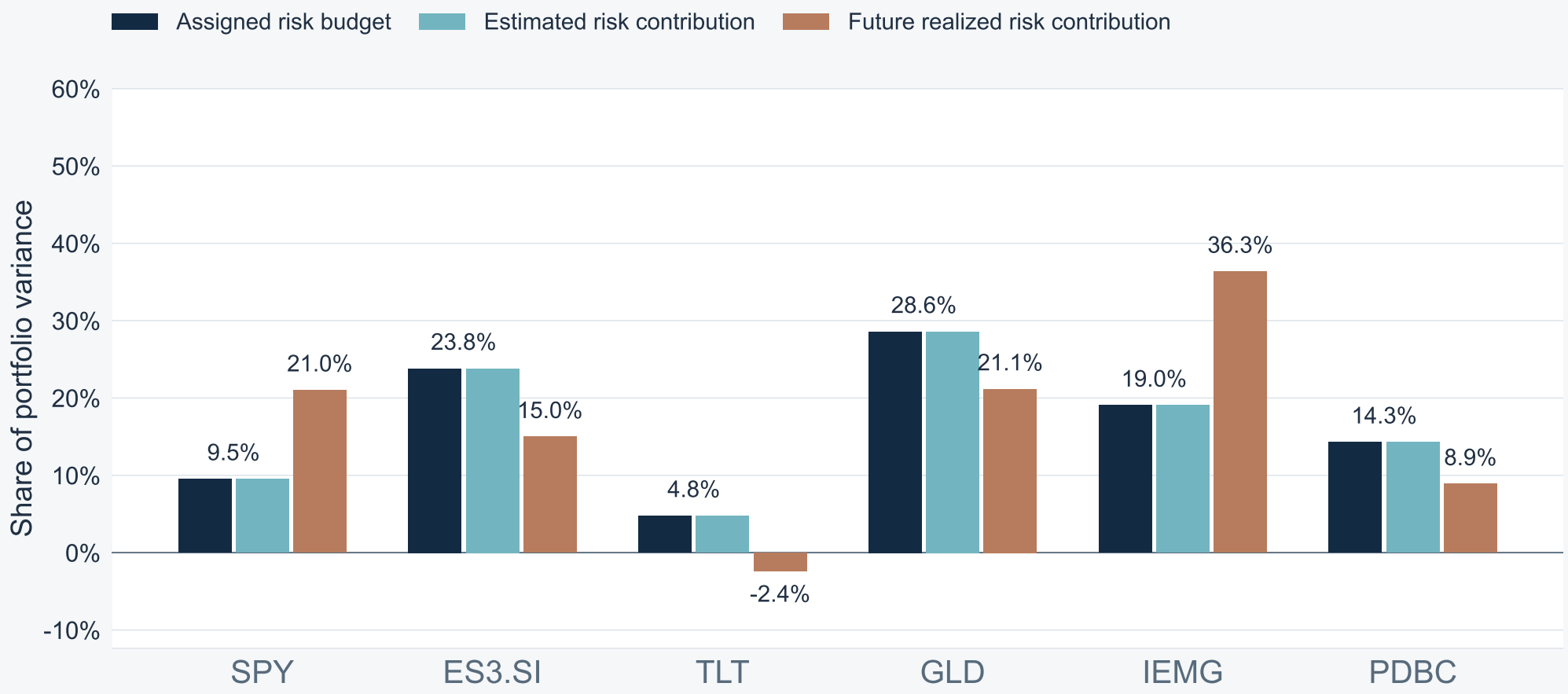
81 complete execution cohorts through Sep 2026 / Sep cohort realized through 2 Oct 2026 / Annualization: sqrt(252)

Estimated risk shares match assigned budgets



Execution: 03 Jan 2024 / Realized through 02 Feb 2024 / Assigned = estimated: one shared label / Signed future shares

Realized risk contributions can be negative



Execution: 03 Feb 2026 / Realized through 03 Mar 2026 / Assigned = estimated: one shared label / Signed future shares

04

Performance Beyond Returns

Compare wealth, risk, trading and capital diversification

Speaker 4

MWRP has competitive recent performance

23.39%

Net compound annual growth rate

2.063

Sharpe ratio, reference rate = 0

–9.15%

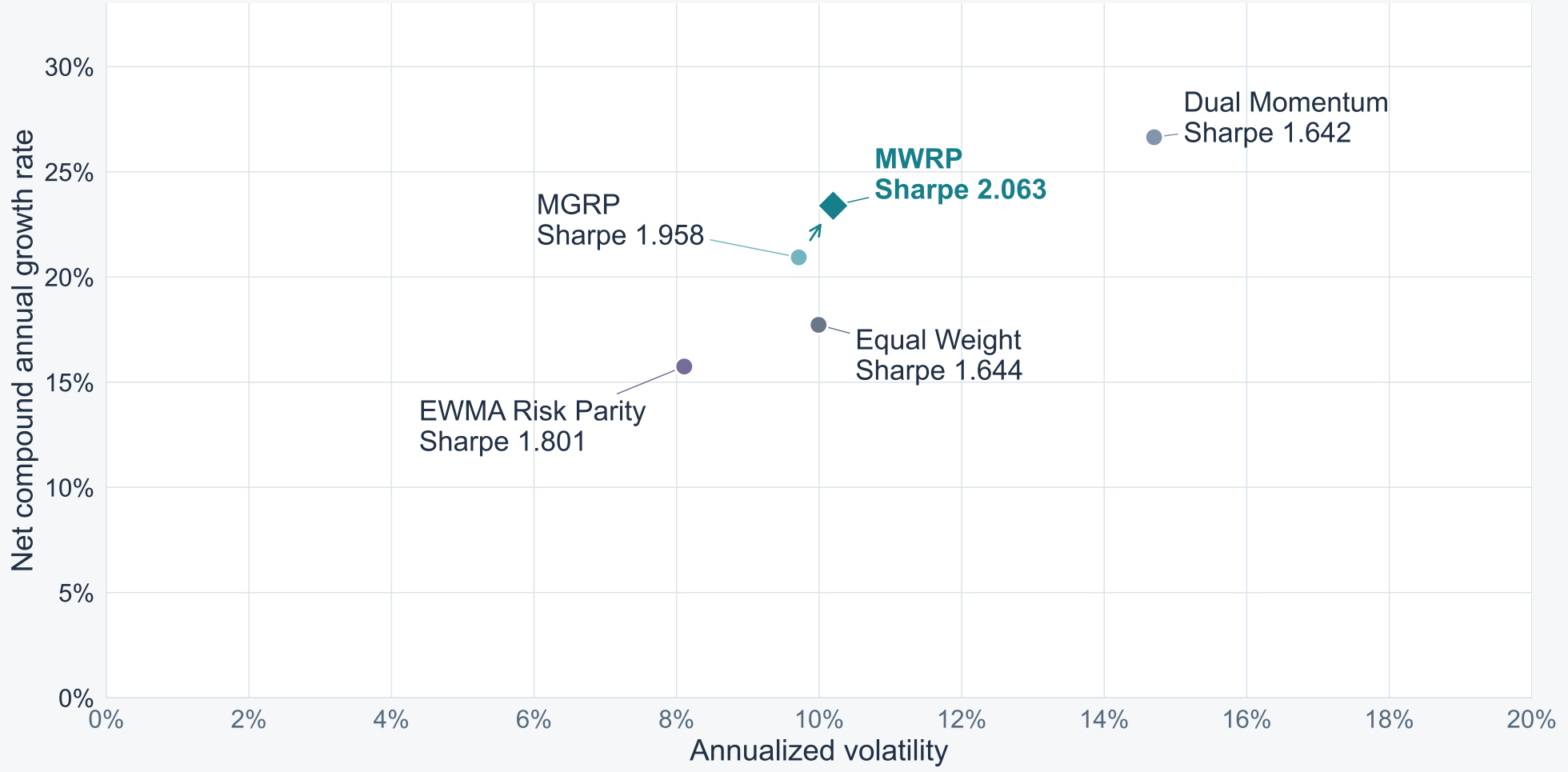
Maximum drawdown (MDD)

Dual Momentum has the higher net CAGR: 26.64%

MWRP has the highest recent Sharpe among the five core strategies.

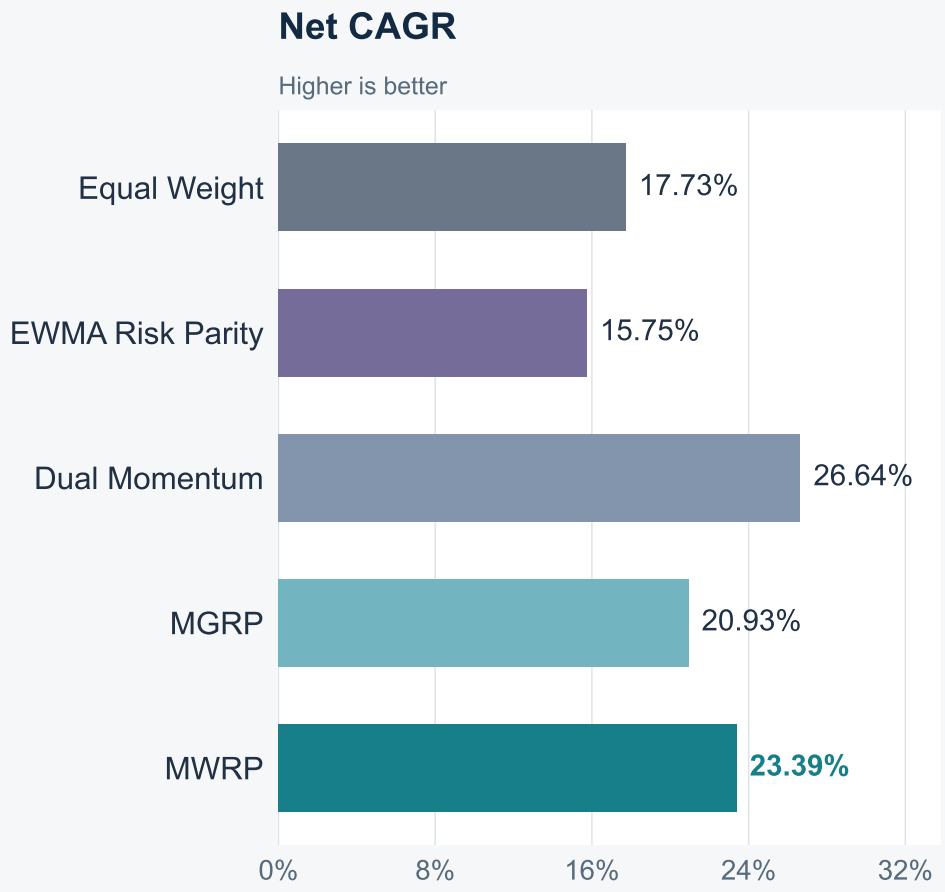
2024 Jan–2026 Sep. Net of 10 basis points (bps) per traded side. 1 bp = 0.01 percentage point.

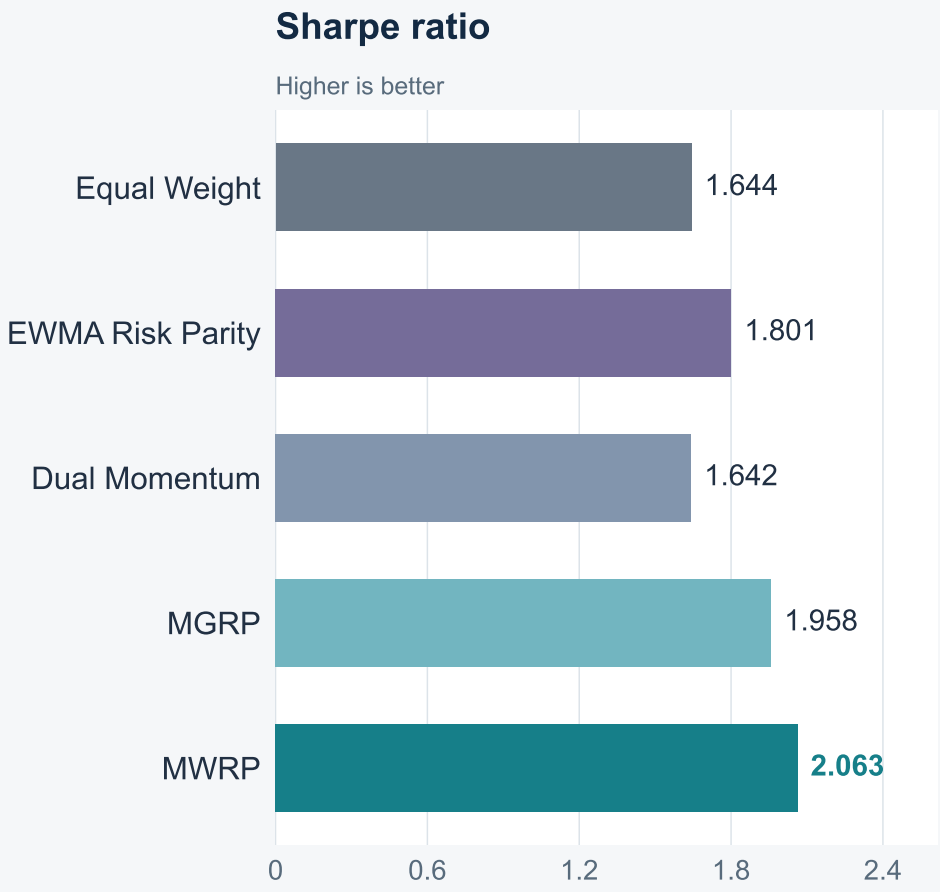
Recent risk-adjusted performance improves with MWRP



2024 Jan-2026 Sep / SGD total return / Net of 10 bps per traded side / Arrow denotes a paired comparison, not an efficient frontier

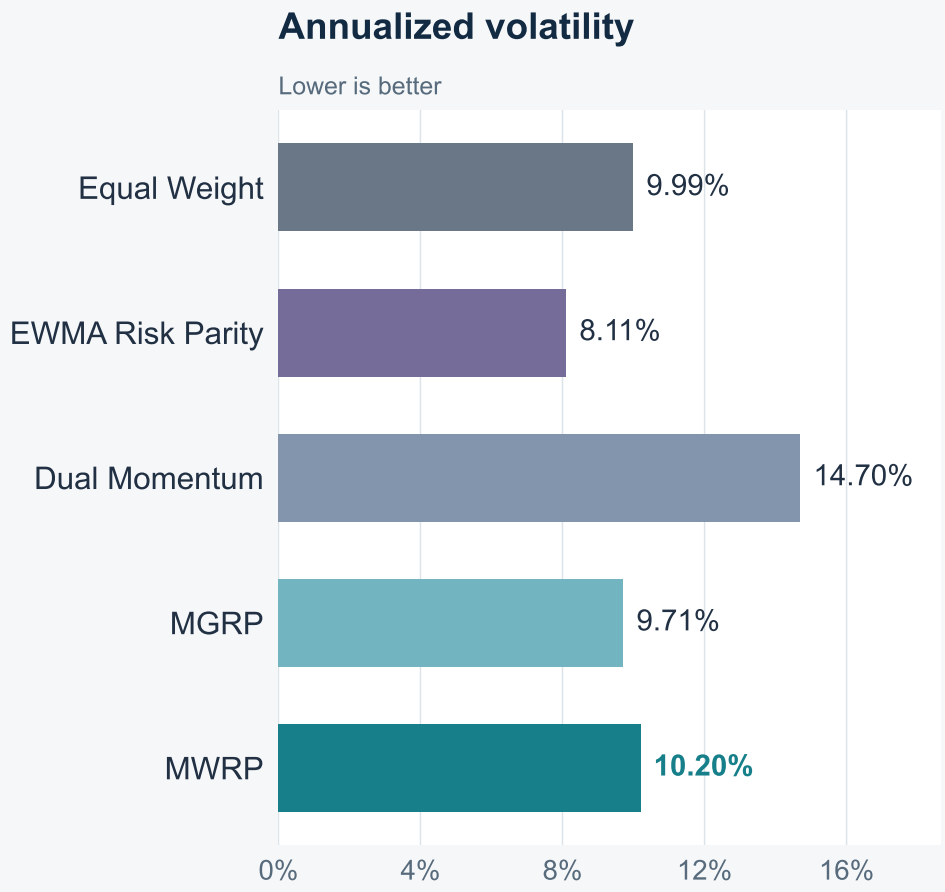
MWRP leads recent Sharpe, Dual leads recent CAGR

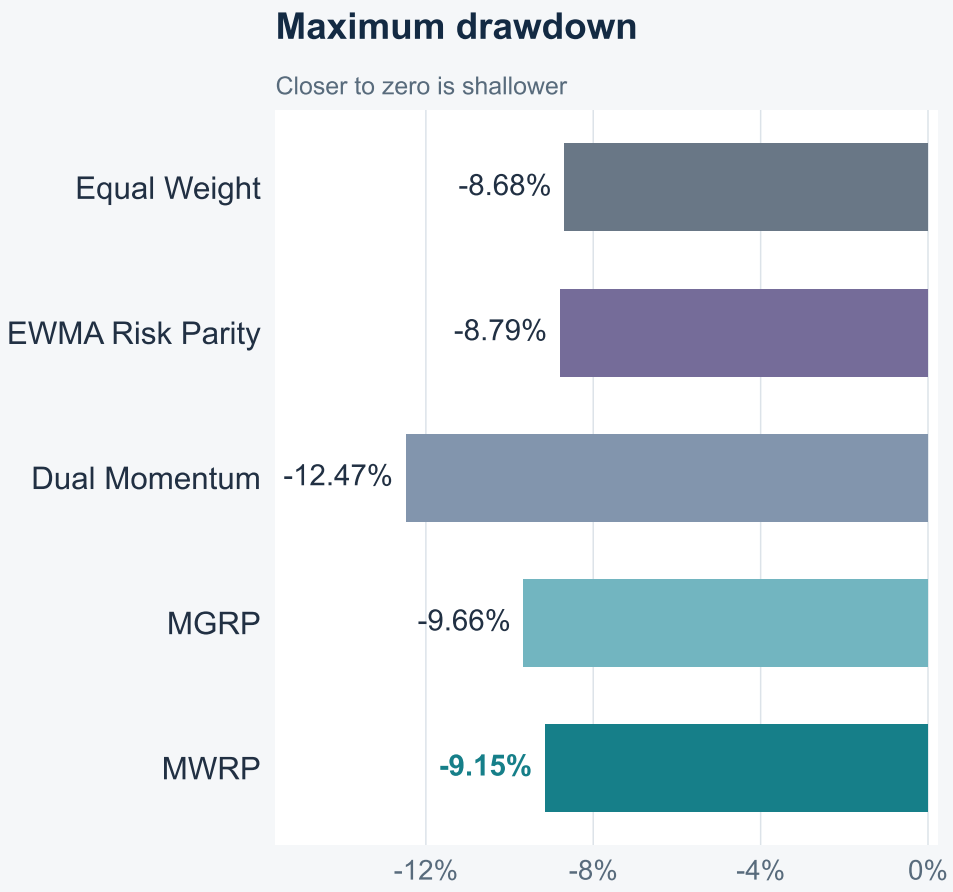




2024 Jan-2026 Sep / SGD total return / Net of 10 bps per traded side / Same strategy order and colors

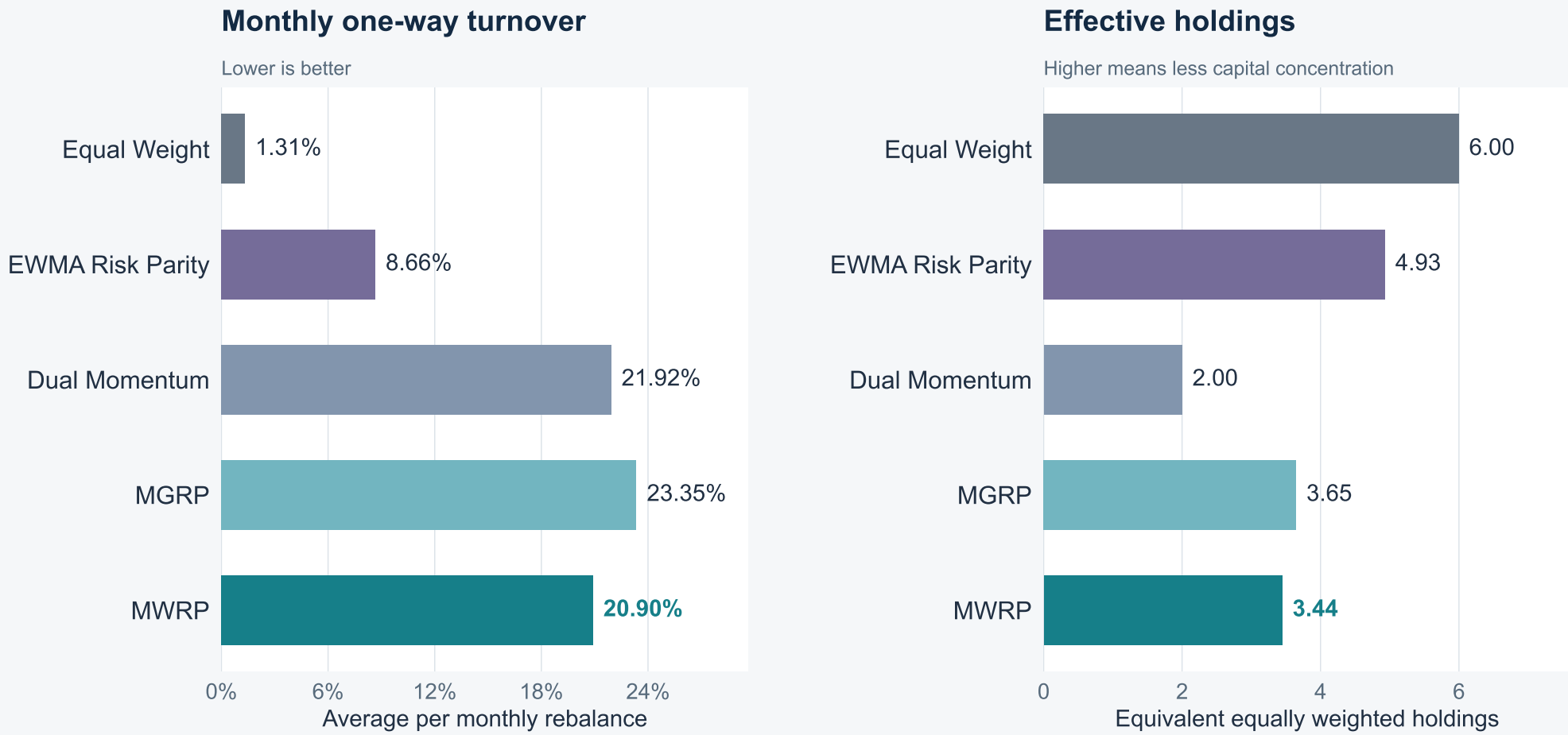
MWRP takes less risk than Dual, but more than MGRP





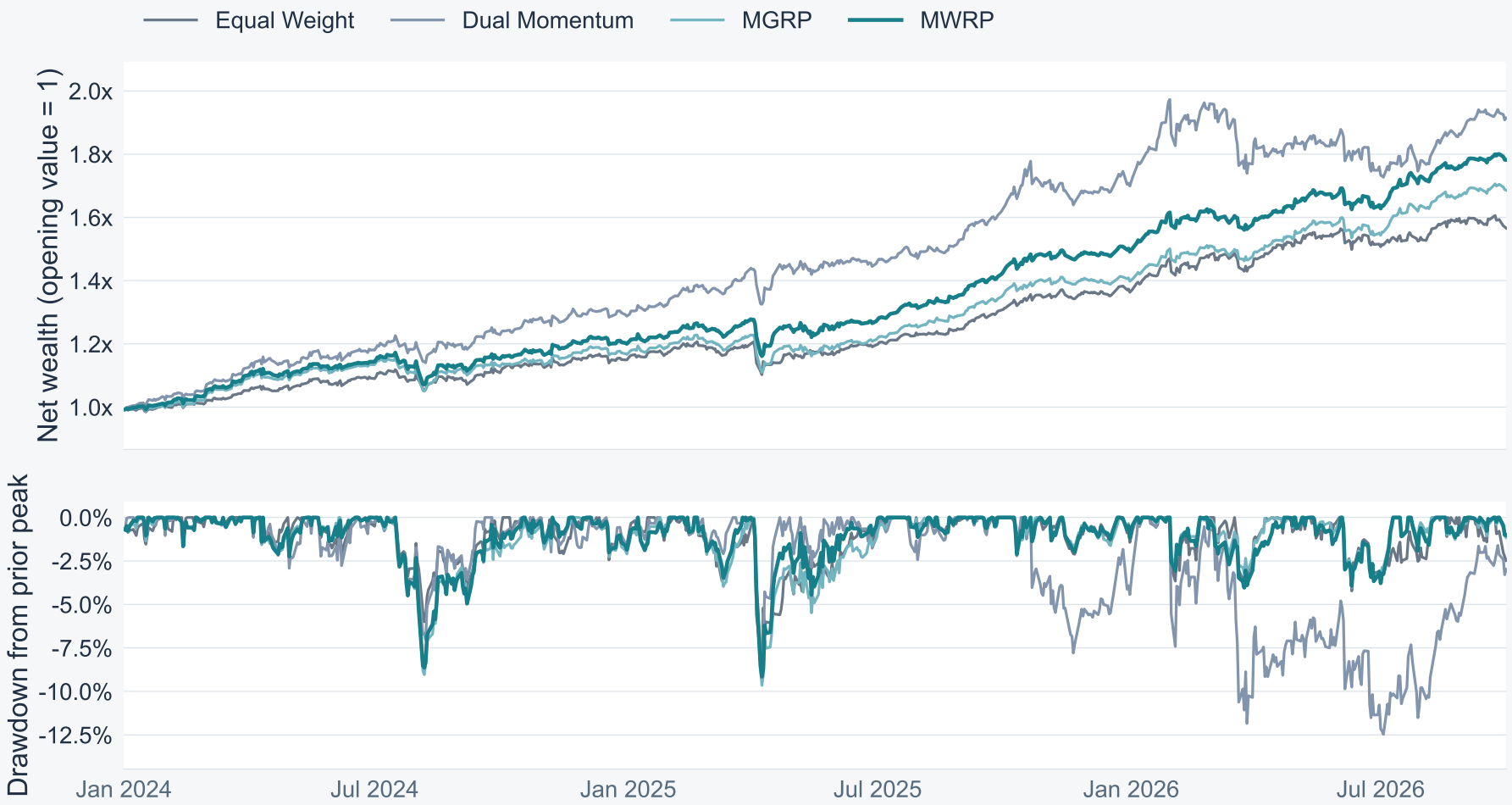
2024 Jan-2026 Sep / SGD total return / Net of 10 bps per traded side / Same strategy order and colors

MWRP balances trading intensity and diversification



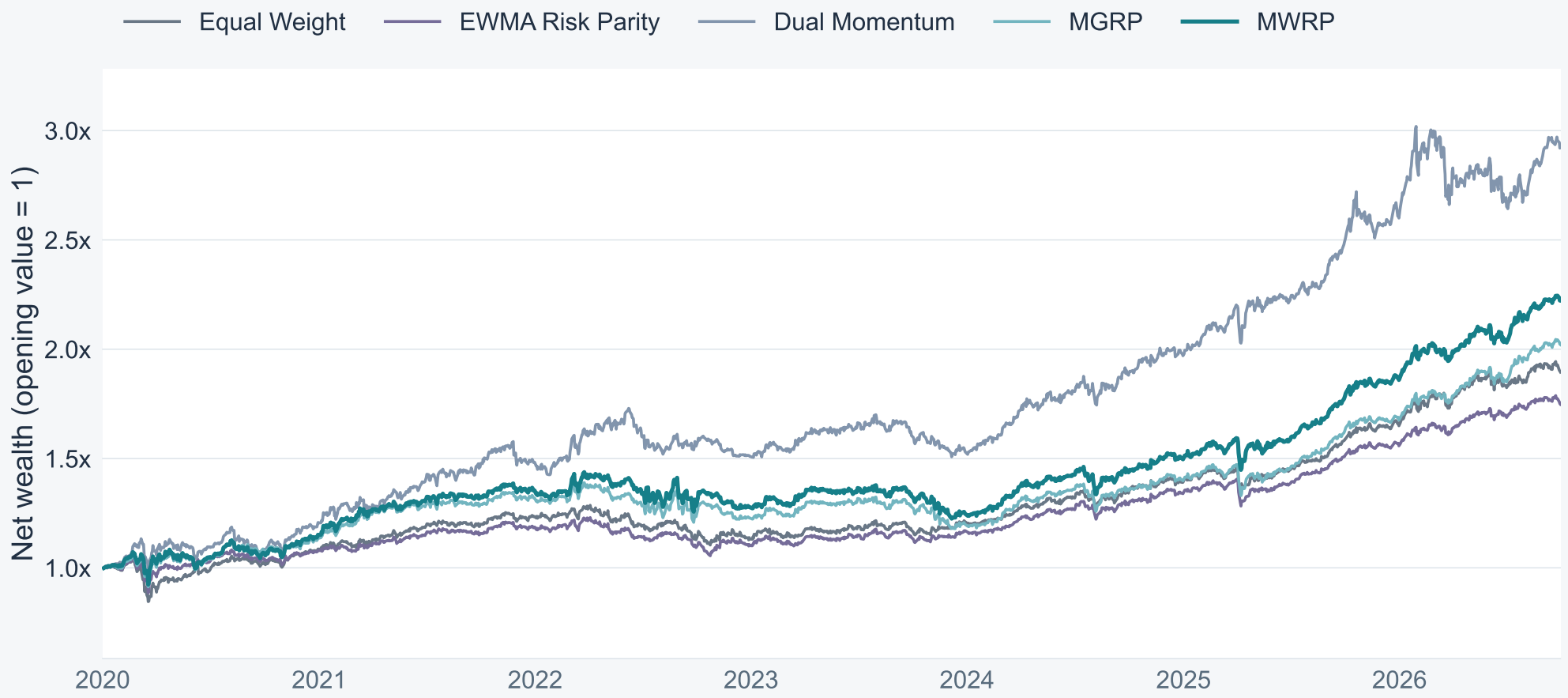
2024 Jan-2026 Sep / SGD total return / Net of 10 bps per traded side / Same strategy order and colors

Recent wealth gains come with different drawdown paths



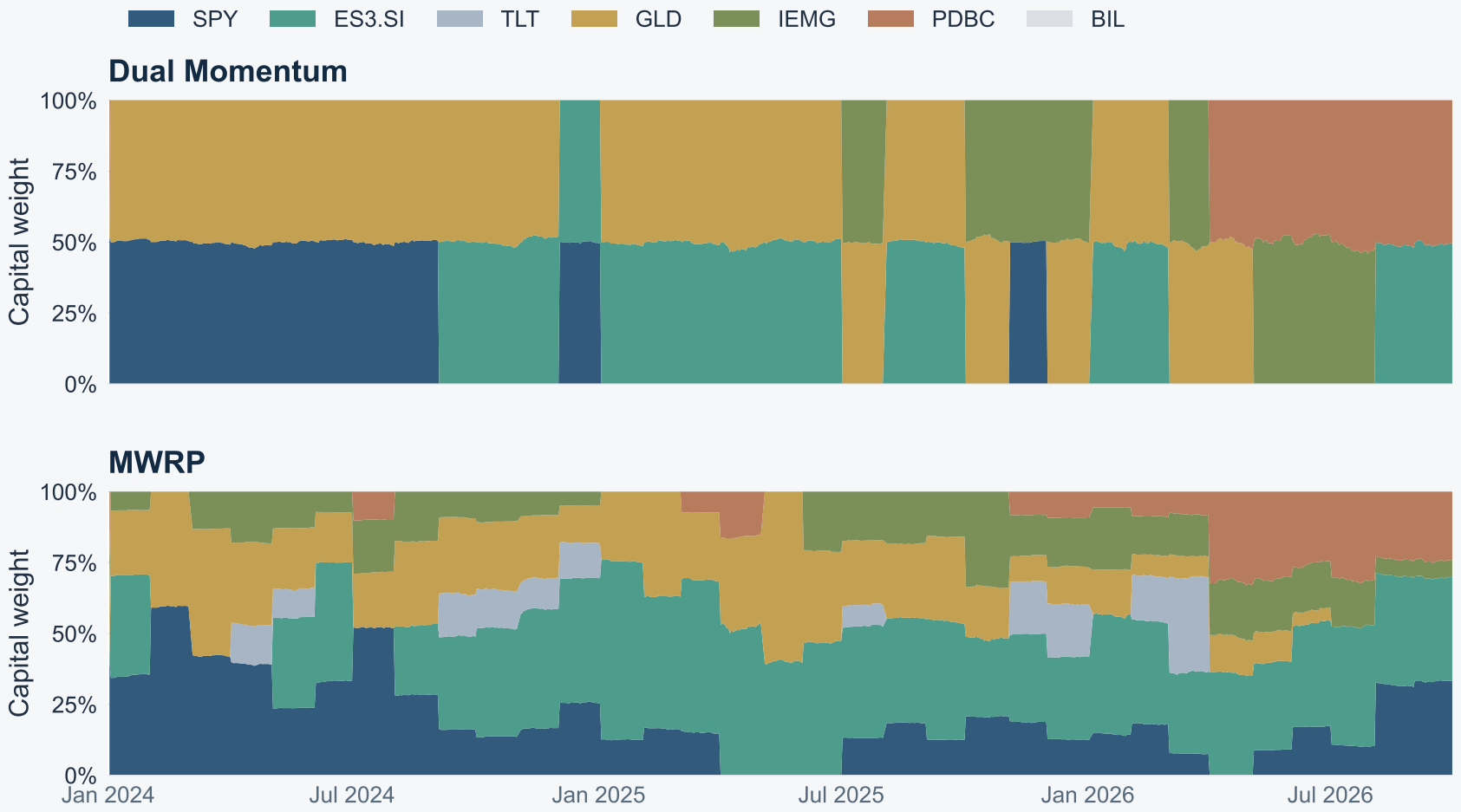
2024 Jan-2026 Sep / SGD total return / Net of 10 bps per traded side / Continuous positions, window-opening wealth normalized to 1

The full history favors Dual in accumulated wealth



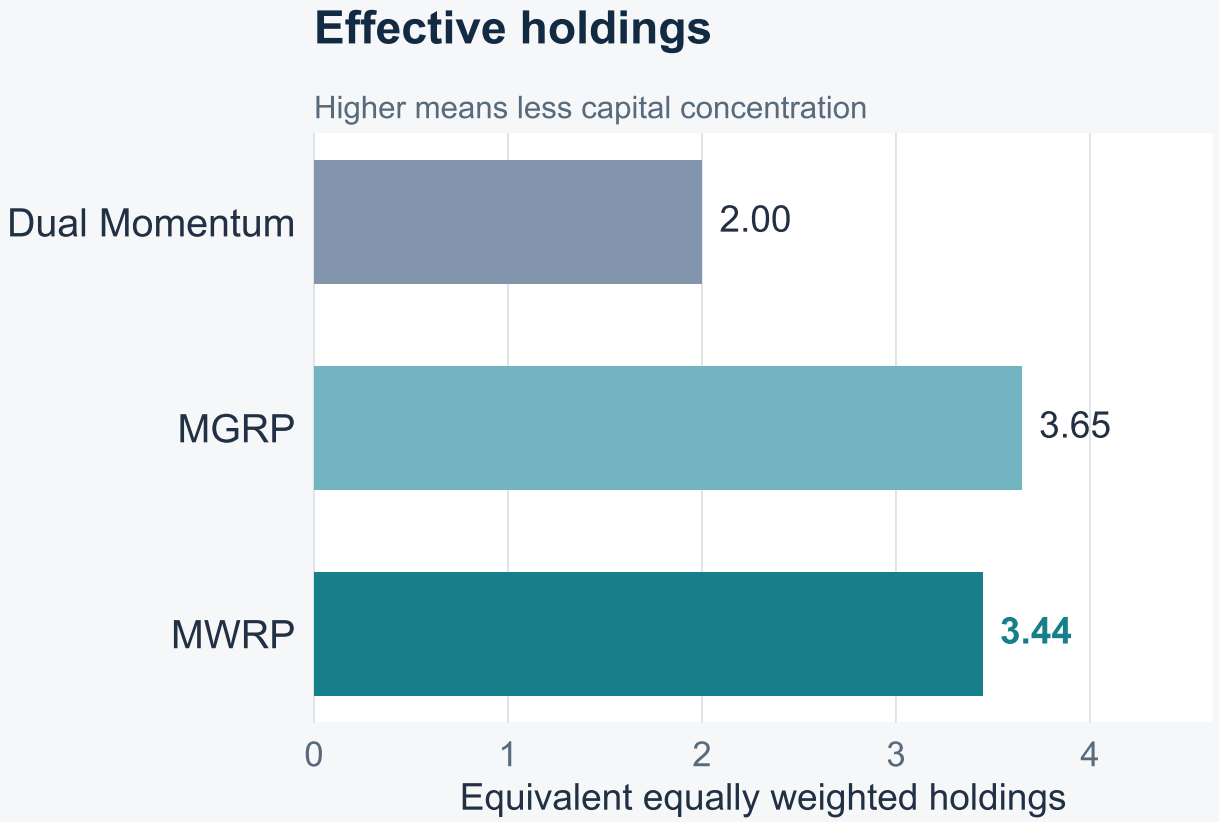
2020 Jan-2026 Sep / Continuous historical net wealth / SGD, 10 bps per traded side / Not a third independent test sample

MWRP distributes capital more broadly than Dual



2024 Jan-2026 Sep / Actual daily close capital weights, including drift / BIL is a USD Treasury-bill ETF, not SGD cash

Diversification is an average, not a position limit



2024 Jan-2026 Sep / Actual drifted portfolio weights

MWRP capital concentration

41.12%

Mean largest actual capital position

63.78%

Peak single-asset weight 2024 Jan–2026 Sep

The same fee rate produces different cumulative costs

Strategy	Fees / opening wealth	Gross minus net CAGR
Dual Momentum	2.26%	0.67 pp
MGRP	1.90%	0.68 pp
MWRP	**1.78%**	**0.62 pp**

Fees depend on actual trades and the wealth path.

CAGR drag is an annualized difference between gross and net wealth paths.

2024 Jan–2026 Sep. Uniform 10 bps per traded side. Cumulative fee amounts are not CAGR differences.

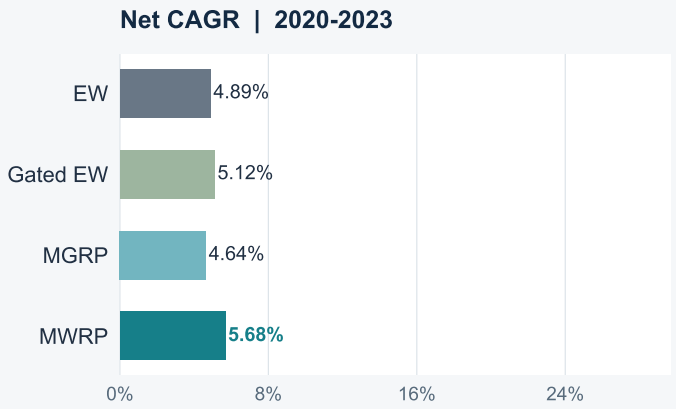
05

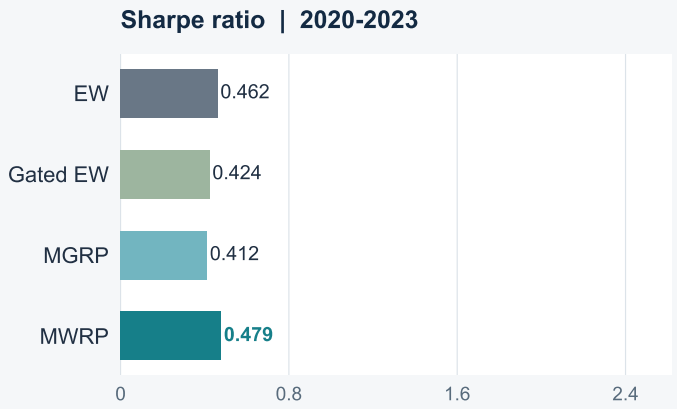
Where Does the Improvement Come From?

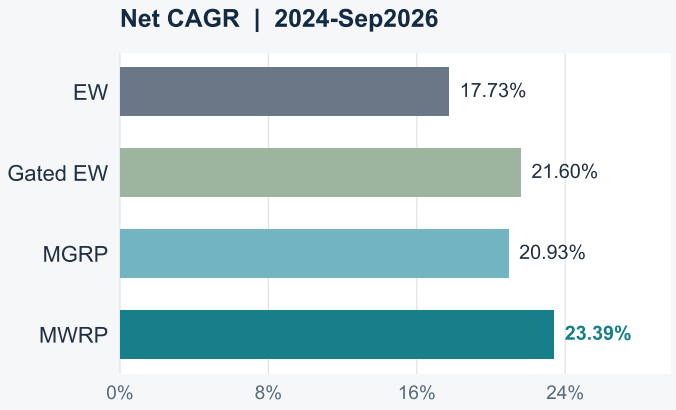
Separate eligibility, allocation and momentum-weighted risk preferences

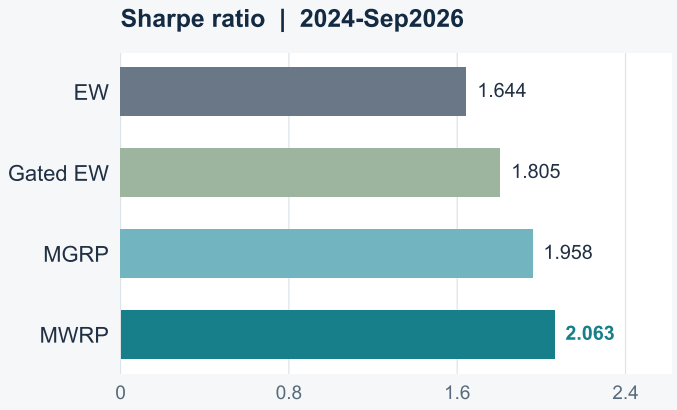
Speaker 5

Eligibility and allocation have distinct historical effects







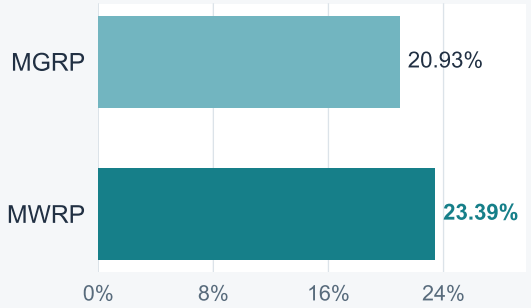


Component comparisons share data, timing and costs / Gated EW versus MGRP changes capital allocation; MGRP versus MWRP changes risk budgets

Momentum-weighted budgets improve on MGRP

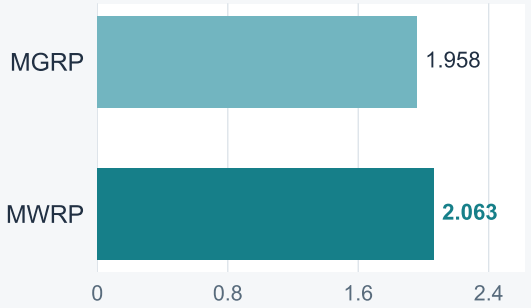
Benefits

Net CAGR



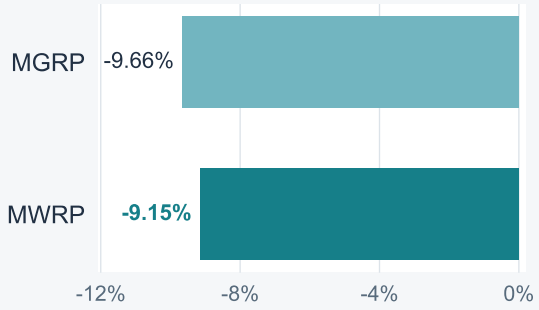
Change: +2.46 pp

Sharpe ratio



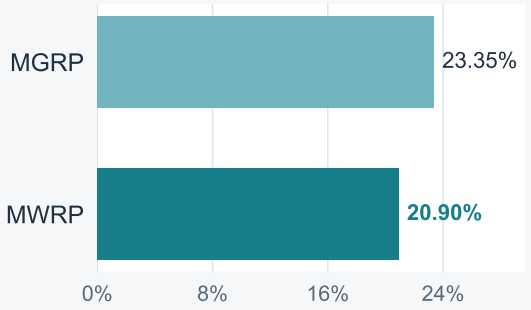
Change: +0.104

Maximum drawdown



Change: +0.50 pp

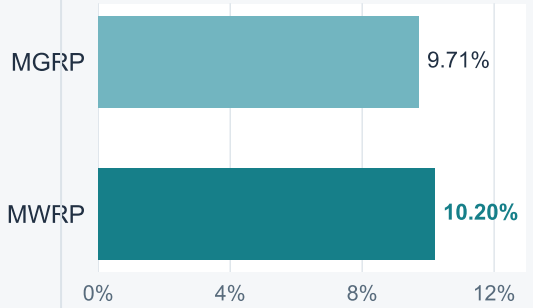
Monthly one-way turnover



Change: -2.44 pp

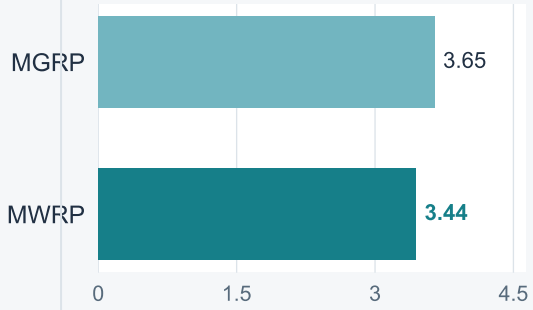
Trade-offs

Annualized volatility



Change: +0.48 pp

Effective holdings



Change: -0.20

2024 Jan-2026 Sep / Separate axis for every metric / Benefits: CAGR, Sharpe, drawdown and turnover; trade-offs: volatility and concentration

Momentum weighting changes participation, not eligibility

01 **Same eligible set and defensive timing**

MGRP and MWRP change allocation, not the list of qualifying assets.

02 **33 of 33 recent months have different target weights**

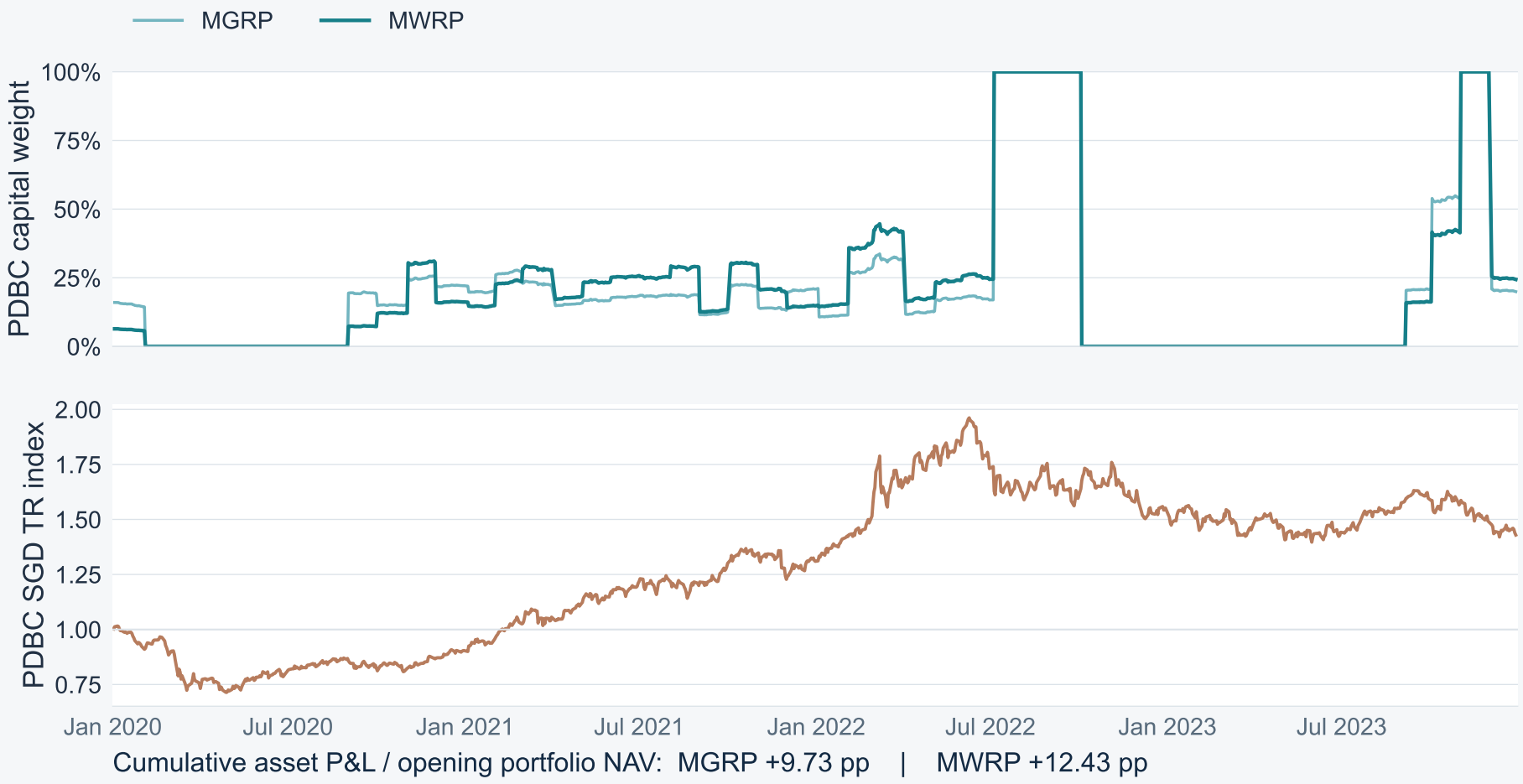
Average allocation distance: 13.18%. This is not turnover.

03 **The strongest momentum asset is largest in capital only 10/33 months**

30.30%. Covariance still determines how risk preferences become money.

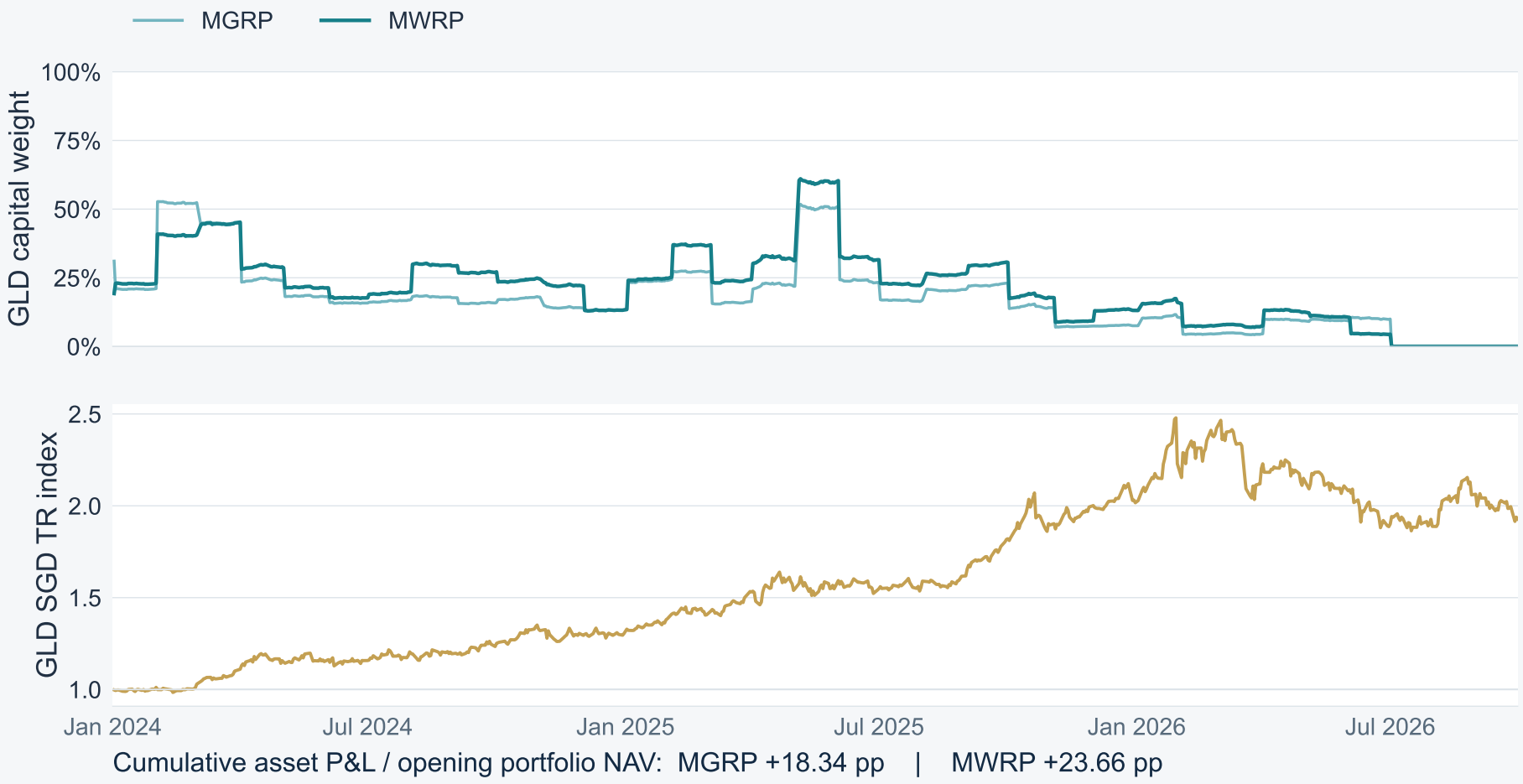
Source: frozen Phase 4 monthly decisions and report. Validation target weights differ in 41/48 months.

MWRP increases participation in the commodity trend



2020 Jan-Dec 2023 / Actual daily exposures / Asset index rebased to 1 at the prior close / Contribution is not CAGR

A larger gold allocation helps explain recent gains



Cumulative asset P&L / opening portfolio NAV: MGRP +18.34 pp | MWRP +23.66 pp

2024 Jan-2026 Sep / Actual daily exposures / Asset index rebased to 1 at the prior close / Contribution is not CAGR

Asset P&L reconciles the improvement in total return

Asset / cost	MGRP	MWRP
SPY	+8.17 pp	+8.14 pp
ES3.SI	+30.77 pp	+33.51 pp
TLT	−2.18 pp	−1.92 pp
GLD	**+18.34 pp**	**+23.66 pp**
IEMG	+13.19 pp	+11.76 pp
PDBC	+2.22 pp	+4.83 pp
BIL	0.00 pp	0.00 pp
Trading costs	−1.90 pp	−1.78 pp
Net total return	**+68.62%**	**+78.20%**

2024 Jan–2026 Sep. Additive P&L (profit and loss) / opening wealth. Not CAGR. Rounded cells may not sum exactly.

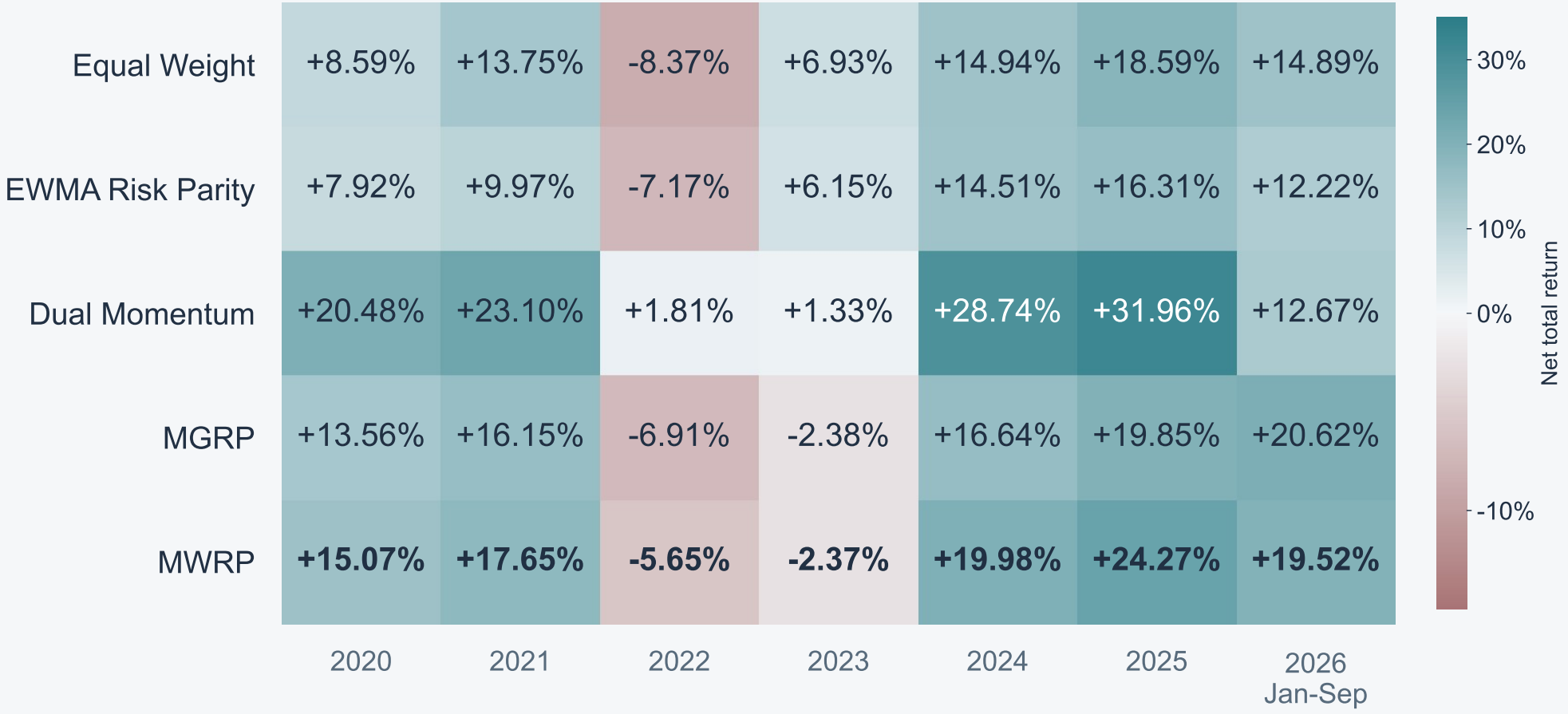
06

Robustness & Final Takeaways

Show less favorable periods and define the limits of the evidence

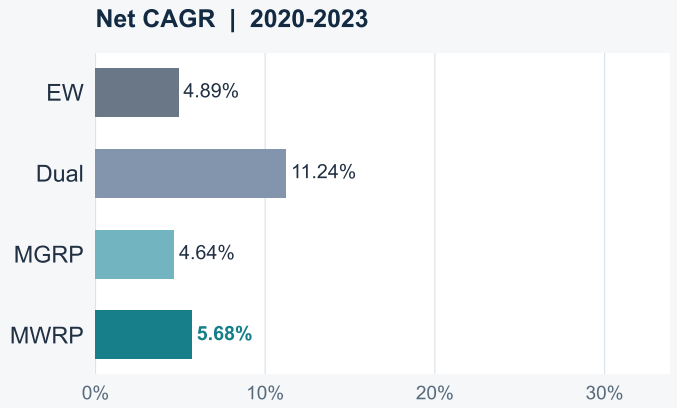
Speaker 1

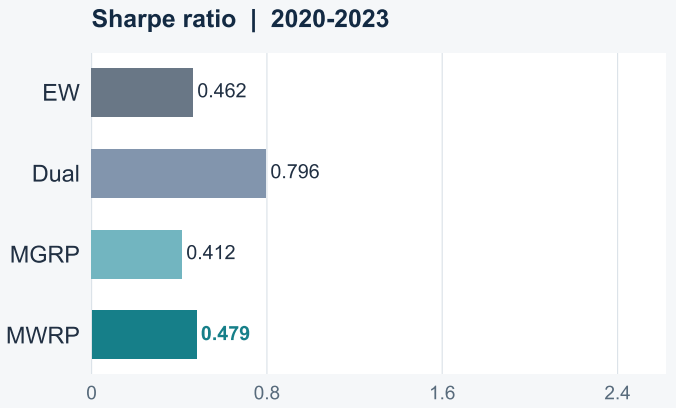
Annual outcomes vary across market environments

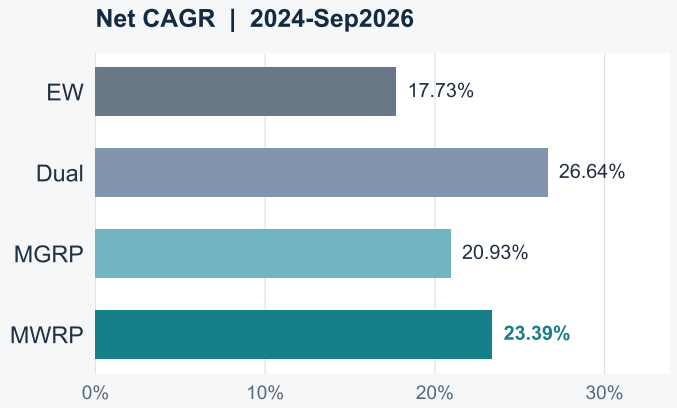


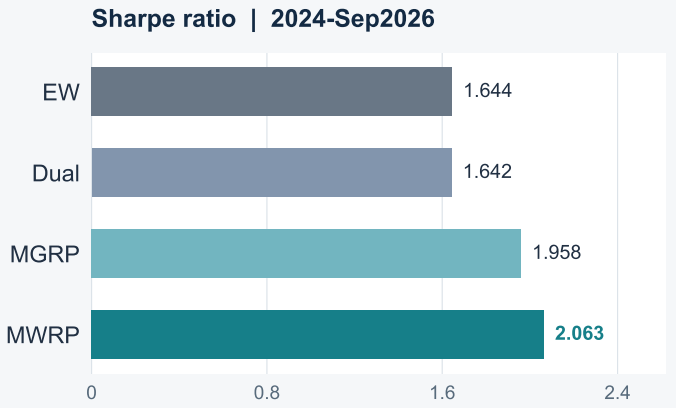
Calendar-year net total returns / 2026 covers Jan-Sep only, not annualized / SGD, 10 bps per traded side

MWRP improves on MGRP across both subperiods



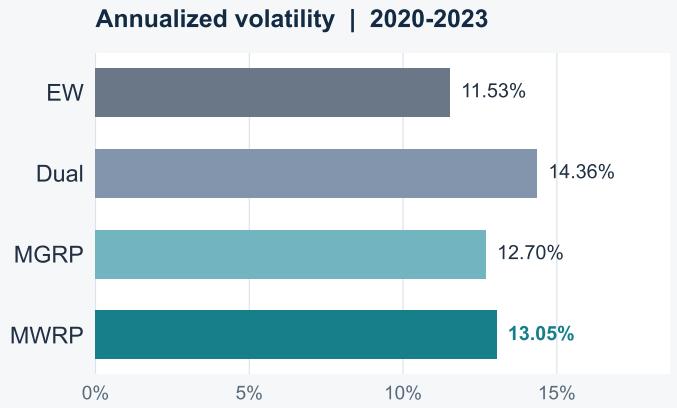


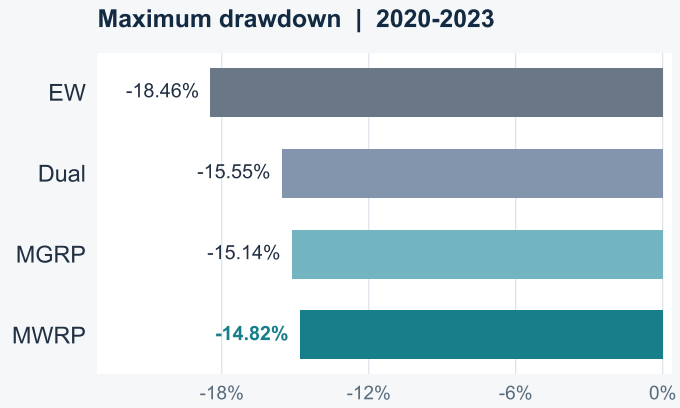


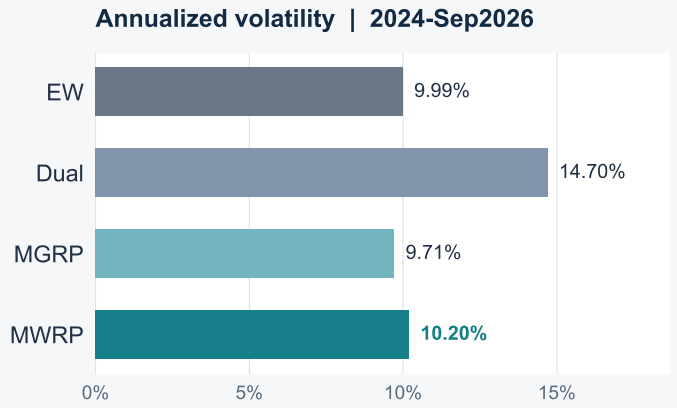


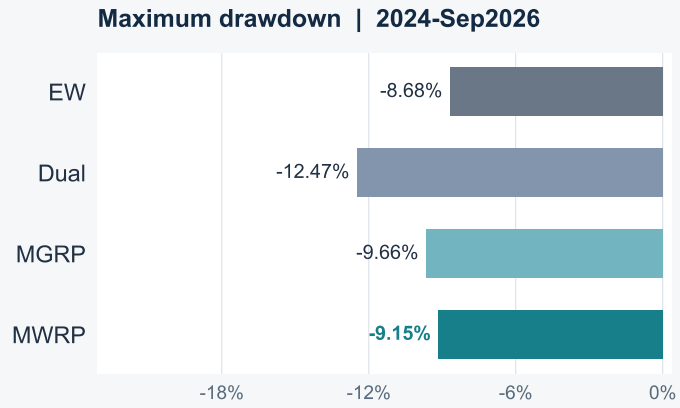
Two non-overlapping historical periods / Earlier results are not independent prospective evidence for a model designed later

Risk and drawdown comparisons also depend on period



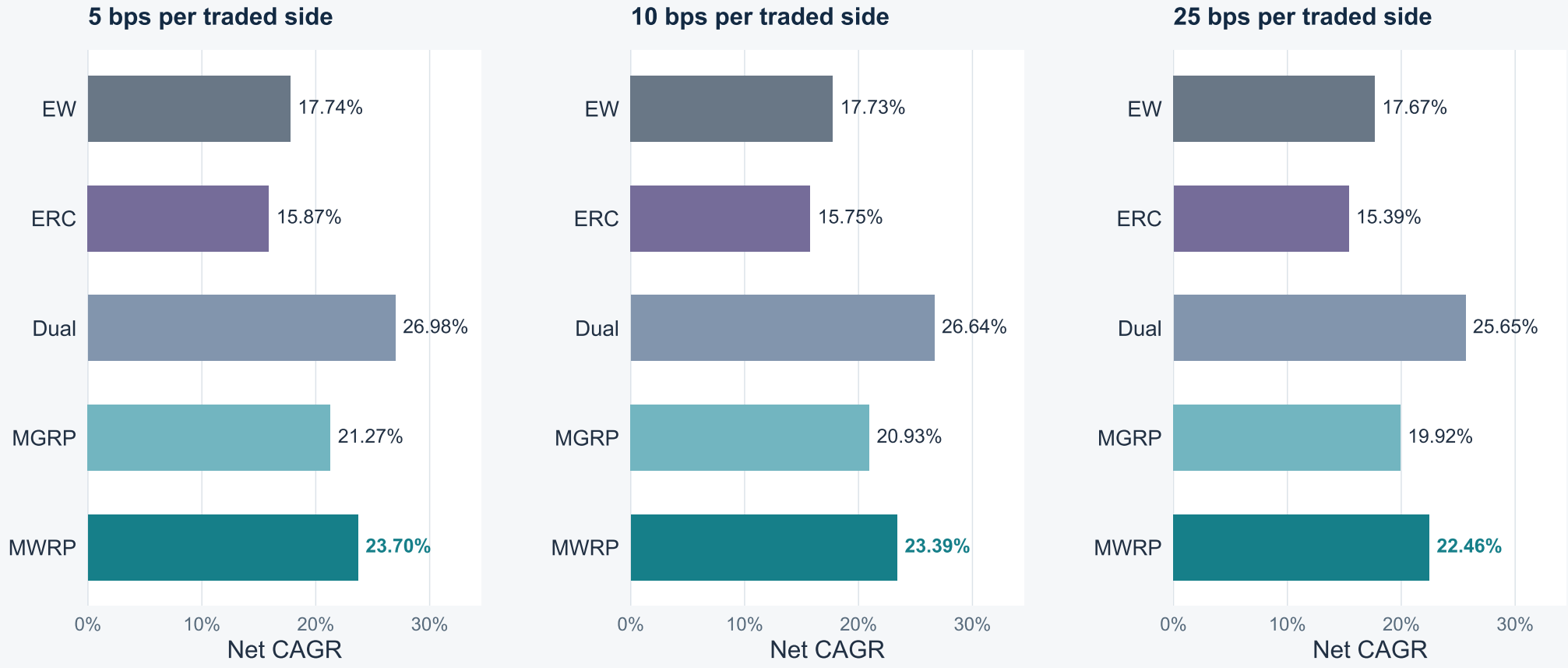






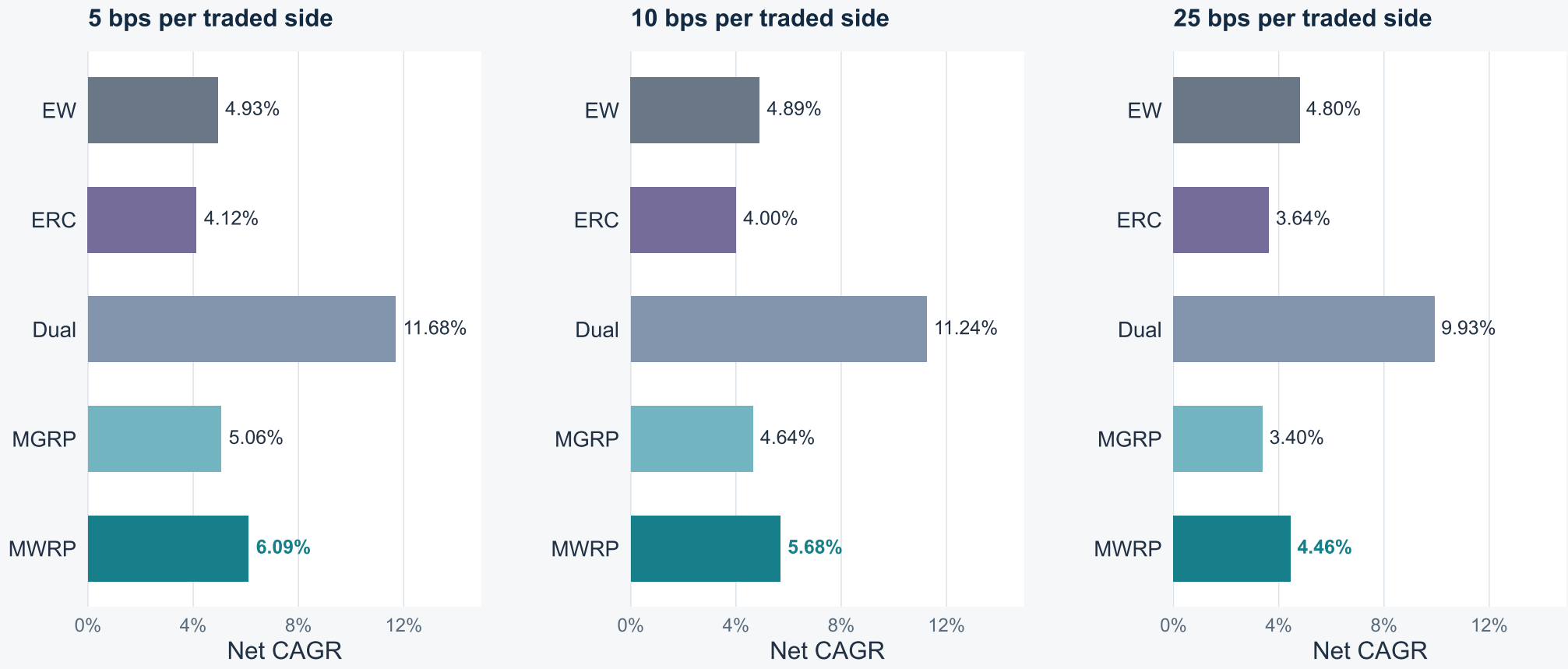
Two non-overlapping historical periods / Earlier results are not independent prospective evidence for a model designed later

Recent findings persist across trading-cost scenarios



2024 Jan-2026 Sep / Same scale in all three panels / EW: Equal Weight; ERC: EWMA Risk Parity / Frozen cost scenarios

Cost sensitivity does not remove the validation gap



2020 Jan-Dec 2023 / Same scale in all three panels / EW: Equal Weight; ERC: EWMA Risk Parity / Frozen cost scenarios

Momentum adds value through the allocation decision

01 **Momentum has economic value in this history**

Concentrated winner selection earns high returns but accepts higher volatility.

02 **Momentum-weighted budgets improve on equal-risk MGRP**

Return, Sharpe and observed drawdown improve in both historical subperiods.

03 **The advantage over Dual depends on the period**

Recent Sharpe leadership does not imply full-history superiority.

Historical evidence leaves material uncertainty

Limitation	Implication
Historical strategy-selection bias	No independent prospective evidence
Limited ETF universe	Results may not generalize to other assets
Asynchronous market and FX closes	Daily basket execution is an approximation
Simplified costs and taxes	Actual investor implementation may differ
Imperfect covariance forecasts	Future risk shares can miss assigned targets
BIL retains USD/SGD risk	Defensive does not mean risk-free in SGD

Next step: freeze the rules and compare future, previously unobserved months.

Momentum informs risk preferences.

Econometrics determines capital allocation.

A more balanced approach to cross-asset momentum investing,
with competitive recent returns and improved recent risk-adjusted performance.

Historical advantages depend on the period. Dual leads full-history CAGR and Sharpe.

REFERENCE

Technical Reference

Definitions, implementation details and complete results

ETF ticker dictionary

Ticker	Official fund name	Quote
SPY	State Street SPDR S&P 500 ETF Trust	USD
ES3.SI	State Street SPDR Straits Times Index ETF	SGD
TLT	iShares 20+ Year Treasury Bond ETF	USD
GLD	SPDR Gold Shares	USD
IEMG	iShares Core MSCI Emerging Markets ETF	USD
PDBC	Invesco Optimum Yield Diversified Commodity Strategy No K-1 ETF	USD
BIL	State Street SPDR Bloomberg 1-3 Month T-Bill ETF	USD

ES3.SI is the vendor symbol. SGX exchange code: ES3. BIL is the defensive asset.

Strategy definitions

Strategy	Eligible set / selection	Allocation
Equal Weight (EW)	All six risk assets	Equal capital
EWMA Risk Parity (ERC)	All six risk assets	Equal forecast risk shares
Dual Momentum	Return Top 2, then compare to BIL	Two 50% capital slots
Gated Equal Weight	All assets with return above BIL	Equal capital among eligible
MGRP	Same eligible set	Equal forecast risk shares
MWRP	Same eligible set	Linear momentum-rank risk shares

MGRP = Momentum-Gated Risk Parity. Frozen source label: MGRP-Base.

For Gated EW, MGRP and MWRP: no eligible assets means 100% BIL. Dual redirects each failed slot.

Growth and maximum drawdown definitions

Compound annual growth rate (CAGR)

$$\mathrm{CAGR} = \left(\frac{V_T}{V_0}\right)^{1/Y} - 1, \qquad Y = \frac{(d_T - d_{\mathrm{start}})_{\mathrm{days}} + 1}{365.25}$$

Maximum drawdown (MDD)

$$\mathrm{MDD} = \min_d \left(\frac{W_d}{\max\{1, W_1, \ldots, W_d\}} - 1\right), \qquad W_d = \frac{V_d}{V_0}$$

V0: continuous opening NAV. VT: ending NAV. W: rebased wealth. Y: actual calendar years. Total return = VT / V0 − 1.

Volatility and Sharpe definitions

Annualized volatility

$$\sigma_{\text{ann}} = \sqrt{252}\, s_r, \qquad s_r^2 = \frac{1}{N-1}\sum_{d=1}^{N}(r_d - \bar{r})^2$$

Sharpe ratio

$$\text{Sharpe} = \sqrt{252}\,\frac{\bar{r}}{s_r}, \qquad R_f = 0$$

r: net daily portfolio return. Mean is arithmetic. s: sample standard deviation, ddof=1. RF=0.

Turnover and effective holdings definitions

Monthly average one-way turnover

$$\overline{\mathrm{TO}}_{1\mathrm{w}} = \frac{1}{M}\sum_{m=1}^{M}\frac{\sum_i |\Delta A_{i,m}|}{2V_m^-}$$

Effective holdings

$$N_{\mathrm{eff},d} = \frac{1}{\sum_{i=1}^{7} w_{i,d}^2}, \qquad \overline{N}_{\mathrm{eff}} = \frac{1}{N}\sum_{d=1}^{N} N_{\mathrm{eff},d}$$

Delta A: actual currency trades. V-minus: pre-fee NAV. w: actual daily close weights, including BIL.

MWRP signal and risk-budget formulation

Six-month spread

$$m_{i,t} = \left(\frac{P_{i,t}^{\mathrm{SGD,TR}}}{P_{i,t-6}^{\mathrm{SGD,TR}}} - 1 \right) - \left(\frac{P_{\mathrm{BIL},t}^{\mathrm{SGD,TR}}}{P_{\mathrm{BIL},t-6}^{\mathrm{SGD,TR}}} - 1 \right)$$

Eligibility

$$E_{i,t} = \mathbf{1}\{m_{i,t} > 0\}\,, \qquad k_t = \sum_{i=1}^{6} E_{i,t}$$

Risk preferences

$$b_{i,t} = \frac{k_t - r_{i,t} + 1}{k_t(k_t+1)/2} \quad (i \in \mathcal{E}_t), \qquad b_{i,t} = 0 \quad (i \notin \mathcal{E}_t)$$

EWMA estimation details

$$Q_d = \lambda Q_{d-1} + (1-\lambda)\varepsilon_d \varepsilon_d^\top, \quad \varepsilon_d = R_d - \bar{R}_{d-1}, \quad \lambda = 0.94,$$

$$\widehat{\Sigma}_d = 0.99 Q_d + 0.01\,\mathrm{diag}(Q_d) + D_d$$

Element	Frozen convention
Initialization	252 valuation returns, sample covariance
Centering	Prior-day expanding mean
Recursion	Raw Q with lambda = 0.94
Stabilization	1% diagonal shrinkage, variance floor 1e-12
Allocation	Eligible risk-asset covariance submatrix

No parameter optimization. Fixed annual covariance scaling leaves normalized capital weights unchanged.

Risk-budget optimization and capital constraints

Optimization

$$x^* = \underset{x_i>0,\ i\in\mathcal{E}_t}{\arg\min}\left\{\tfrac{1}{2}x^\top\widehat{\Sigma}_t x - \sum\nolimits_{i\in\mathcal{E}_t} b_{i,t}\log x_i\right\},$$

$$w_i = x_i^* \Big/ \sum\nolimits_{j\in\mathcal{E}_t} x_j^*$$

Capital constraints and fallback

$$w_i \geq 0, \qquad \sum_{i=1}^{7} w_i = 1,$$

$$k_t = 0 \Rightarrow w_{\mathrm{BIL}} = 1, \qquad k_t > 0 \Rightarrow w_{\mathrm{BIL}} = 0$$

Estimated risk-contribution formula: Slide 12. No leverage. Eligible sleeve only. k=1: sole asset receives 100%.

Realized risk and P&L use different decompositions

Signed realized risk shares

$$c_{i,d} = w_{i,d-1} R_{i,d}, \qquad q_{i,m} = \frac{\mathrm{Cov}_{d \in H_m}(c_{i,d}, \sum_j c_{j,d})}{\mathrm{Var}_{d \in H_m}(\sum_j c_{j,d})}$$

Additive profit and loss (P&L)

$$C_i = \frac{\sum_{d \in \mathcal{D}} A_{i,d-1} R_{i,d}}{V_0}, \qquad \sum_i C_i + C_{\text{fees}} = \frac{V_T}{V_0} - 1$$

H: exact (entry close, next close] interval. c: gross asset return contribution. Fees excluded from risk, included in P&L.

Complete performance: recent evaluation

Strategy	Net CAGR	Sharpe	Ann. vol	MDD	Monthly turnover	Effective holdings
Equal Weight	17.73%	1.644	9.99%	-8.68%	1.31%	6.00
EWMA Risk Parity	15.75%	1.801	8.11%	-8.79%	8.66%	4.93
Dual Momentum	26.64%	1.642	14.70%	-12.47%	21.92%	2.00
Gated Equal Weight	21.60%	1.805	10.90%	-8.65%	18.44%	4.24
MGRP	20.93%	1.958	9.71%	-9.66%	23.35%	3.65
MWRP	23.39%	2.063	10.20%	-9.15%	20.90%	3.44

2024 Jan-2026 Sep / SGD / Net, 10 bps per side / Effective holdings calculated from actual daily close weights including BIL

Complete performance: historical validation

Strategy	Net CAGR	Sharpe	Ann. vol	MDD	Monthly turnover	Effective holdings
Equal Weight	4.89%	0.462	11.53%	-18.46%	2.55%	6.00
EWMA Risk Parity	4.00%	0.450	9.52%	-15.20%	9.63%	4.94
Dual Momentum	11.24%	0.796	14.36%	-15.55%	32.84%	1.96
Gated Equal Weight	5.12%	0.424	13.70%	-15.78%	31.32%	3.28
MGRP	4.64%	0.412	12.70%	-15.14%	33.21%	2.99
MWRP	5.68%	0.479	13.05%	-14.82%	32.35%	2.85

2020 Jan-Dec 2023 / SGD / Net, 10 bps per side / Effective holdings calculated from actual daily close weights including BIL

Complete performance: continuous full history

Strategy	Net CAGR	Sharpe	Ann. vol	MDD	Monthly turnover	Effective holdings
Equal Weight	9.94%	0.902	10.93%	-18.46%	2.05%	6.00
EWMA Risk Parity	8.63%	0.946	8.98%	-15.20%	9.23%	4.93
Dual Momentum	17.26%	1.146	14.50%	-15.55%	28.40%	1.97
Gated Equal Weight	11.54%	0.908	12.63%	-15.78%	26.08%	3.67
MGRP	10.99%	0.938	11.58%	-15.14%	29.19%	3.26
MWRP	12.56%	1.025	11.97%	-14.82%	27.68%	3.09

2020 Jan-2026 Sep / SGD / Net, 10 bps per side / Effective holdings calculated from actual daily close weights including BIL

Trading assumptions and evidence boundaries

Item	Convention
Signal information	Month-end history, available next day at 06:00 UTC
Execution	Second common fresh-quote trading-day close basket
Returns	Old positions through replacement close, new positions thereafter
Main trading cost	10 bps per traded side on actual currency trades
Additional scenarios	5 and 25 bps, all other rules fixed
Fund expenses	Already reflected in adjusted prices
Not explicitly modeled	Investor taxes, separate FX fees, time-varying impact

Retrospective simulation. Asynchronous market closes remain a material approximation.

Group member contributions

Member	Presentation responsibility	Actual contribution
[Member 1]	Sections 01 and 06	[To be confirmed]
[Member 2]	Section 02	[To be confirmed]
[Member 3]	Section 03	[To be confirmed]
[Member 4]	Section 04	[To be confirmed]
[Member 5]	Section 05	[To be confirmed]

Presentation responsibilities do not establish who performed the research.